Hochschule der Bayerischen Wirtschaft

M A S T E R T H E S I S

Understanding Risk Awareness and Decision Making in Early-Stage Project Planning: Comparative Study of Project Managers Across Industries

submitted by:

Devarshi Kansara

(student number: 200445)

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Supervisor: Prof. Dominik Bösl (HDBW)

HDBW Hochschule der Bayerischen Wirtschaft

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Devarshi Kansara

München, 09.02.2026

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Abstract

This master's thesis examines the pivotal beginning period of the project lifecycle, referred to as the first 20 percent, where about 70 percent of project failures occur due to decisions made with insufficient information. The study concentrates on the physically irreversible domains of Construction and Manufacturing, intentionally omitting the IT sector because of its absence of material permanence. The research employs semi-structured interviews with 24 practitioners from Germany and India to analyze the Universal PM DNA, uncovering that seasoned managers frequently circumvent formal risk registers in favour of somatic verification. This entails employing actual site visits and sensory intuition to uncover the objective reality that digital dashboards frequently obscure as courteous fabrications. The analysis reveals a significant trust gap, with 93 percent of managers expressing scepticism in entirely digital data, and 90 percent demonstrating a strategic redundancy reflex by voluntarily incurring a financial safety premium to secure assurance and mitigate the psychological distress associated with project delays. The data reveal two unique archetypes: the German Process Guardian, who employs strict documentation as a bureaucratic shield to mitigate legal liability in a contentious setting. The Indian Resource Navigator, on the other hand, uses creative problem-solving to deal with environmental turmoil and a lack of resources in the supply chain. The study also points to a big discrepancy in experience: junior managers are very aware of small technical mistakes, while senior veterans are more concerned with big strategic dangers. This creates significant blind spots when it comes to physical issues. In conclusion, the thesis posits that enhancing project success rates necessitates transcending dependence on software towards a human-centric methodology characterized by perspective pairing, which integrates the technical acumen of junior personnel with the legal expertise of senior practitioners. The last suggestions call for reverse training, in which German managers learn how to adapt to chaos and Indian managers learn how to standardize digital processes. They also call for the use of truth link technologies that give managers the real-time physical verification they need. This research ultimately shows that good early-stage planning isn't about getting rid of uncertainty with data; it's about giving managers the mental and structural strength they need to connect the digital plan to the real world.

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List of abbreviations

PM: Project Manager

PRM: Project Risk Management

PMP: Project Management Professional (referenced in the context of certified managers)

SME: Small and Medium-sized Enterprises

SaaS: Software as a Service

Mfg: Manufacturing

BIM: Building Information Modelling

FMEA: Failure Mode and Effects Analysis

HAZOP: Hazard and Operability

API: Application Programming Interface

UI: User Interface

NDM: Naturalistic Decision Making

RPD: Recognition-Primed Decision

CYA: Cover Your Ass (referenced as a defensive decision-making strategy)

FDA: Food and Drug Administration (implied context: regulatory compliance)

EU MDR: European Union Medical Device Regulation

GDPR: General Data Protection Regulation (implied context: compliance)

BaFin: Federal Financial Supervisory Authority (implied context: German financial regulation)

DORA: Digital Operational Resilience Act

VOB: Vergabe- und Vertragsordnung für Bauleistungen (referenced as a formal tool/shield in German construction)

1 Introduction

1.1 Early-stage project planning as a critical phase

Project management is the strategic framework that helps a project go from idea to completion. It's not just a list of things to do; it's a methodical way to bring together people, resources, and risk-reduction plans to reach specific goals. In digital business settings where the market is always changing and technology is limited, project management is an important way to make sure that time and resources are used wisely to reach goals. It is the process that turns abstract business ideas into real-life situations.

1.1.1 Stages of Project Management

The initial planning phase serves as the definitive baseline for the entire project lifecycle. Within the digital business sector, this stage involves significant front-loading, during which critical decisions about scope, resource allocation, and risk architecture are determined. Empirical evidence indicates that the stakes are considerable; industry research suggests that nearly 70% of project failures originate from inadequate initial planning. This trend is largely attributable to the reality that, although expenditures are incurred during execution, approximately 80% of total project costs are committed during these early conceptual decisions.

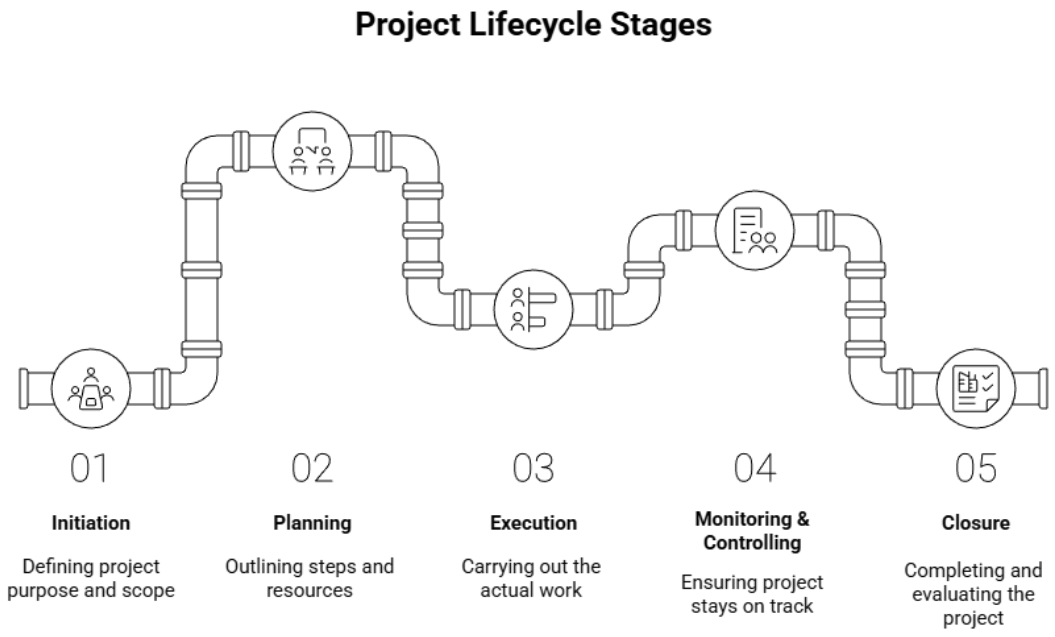


Figure 1: 5 Stages of project life cycle

As a result, deviations or miscalculations during this phase can trigger cascading effects, such as scope creep and resource fragmentation. If not addressed promptly, these issues may develop into systemic failures that require costly, reactive interventions in later development stages. Establishing rigorous baselines and achieving stakeholder alignment from the outset enables a project to transition from reactive management to proactive governance, thereby increasing the likelihood that the final output remains viable in a dynamic digital market.

1.2 Role of risk in early-stage project planning

Risks play a central role in early-stage project planning because they represent uncertainties that can derail goals, timelines, and budgets if not addressed upfront. Project managers must identify, assess, and plan responses to these risks during this phase to establish realistic baselines and avoid costly changes later. Effective early risk handling separates successful projects from those that face constant crises, as unaddressed risks often account for 30-50% of total project disruptions.

- Early attention to uncertainty allows teams to shape project conditions before commitments become rigid.
- Discussing potential obstacles early supports more realistic promises to stakeholders.
- Considering downside scenarios at the beginning encourages more flexible plans and buffers.
- Documenting possible negative events early helps create a shared understanding of what could go wrong.
- Talking about uncertainty from the start promotes a culture where issues are raised sooner rather than hidden.

1.3 Risk awareness and decision making by project managers

Risk awareness and decision making by project managers form the core of early-stage success. Project managers with high-risk awareness spot potential issues early through experience, intuition, and structured methods, while effective decision making under uncertainty balances data, stakeholder input, and practical judgement to prioritise actions. These skills determine whether risks become manageable challenges or project-threatening crises, as studies show experienced managers reduce failure rates by 20-30% through better early decisions.

1.4 Industry specific nature of risk

Risks in projects take on an industry-specific nature because each sector faces unique uncertainties shaped by its environment, regulations, and operations. Construction managers

prioritise weather delays, safety hazards, and permit issues; IT professionals focus on technical complexity, changing requirements, and cybersecurity; while manufacturing leaders address supply chain disruptions, quality control, and equipment failures. This variation means universal risk approaches often fail, as industry context determines both which risks appear first and how they impact project outcomes. Sector characteristics create distinct patterns of uncertainty that managers learn to anticipate.

- Environmental factors influence which challenges surface first in project lifecycles.
- Operational realities shape the types of obstacles teams routinely encounter.
- Regulatory environments determine common compliance hurdles across projects.
- Experience within a field builds intuition for predictable versus unexpected threats.

1.5 Limitations of Existing Research

1. Lack of Cross-Industry Comparative Frameworks

- Most studies focus on individual industries, such as construction, IT, or manufacturing, rather than systematically comparing differences in risk awareness across sectors.
- Although current literature often treats project risk identification as a universal process, industry context influences which risks managers prioritize. This empirical gap remains largely unexamined.
- There is a lack of structured research explaining why project managers from different industries develop distinct mental models of risk during early planning, which can create blind spots in cross-functional work.

2. Insufficient Focus on the First 20% of Project Duration

- Although research indicates that 70% of project failures result from inadequate early planning, most risk management studies focus on the entire project lifecycle rather than the critical initial phase.
- There is limited empirical data on how risk identification and decision-making patterns differ between early-stage planning (first 20% of project duration) and subsequent phases.
- As a result of this temporal gap, practitioners lack evidence-based guidance tailored to early-stage risk awareness practices.

3. Cognitive Biases in Risk Identification Remain Underexplored

- Research identifies "unknown unknowns" and cognitive biases as central factors in incomplete risk identification, yet it does not examine whether these blind spots differ systematically across industries or project manager backgrounds.
- Practitioners face imagination and experience gaps that hinder recognition of industry-

atypical risks, but current frameworks neither quantify this effect nor provide guidance for improvement.

- The relationship between cognitive biases, industry experience, and the quality of early-stage decision-making remains largely unresearched.

4. Weak Empirical Evidence on Actual Decision-Making Under Uncertainty

- Most studies describe theoretical risk processes, such as identification, analysis, and response, but rarely capture real-time data on how project managers make decisions when confronted with multiple uncertainties and incomplete information during early planning.
- There is limited qualitative evidence from experienced managers regarding the decision heuristics, intuition, and structured methods employed during high-uncertainty early project phases.
- No comparative studies demonstrate which decision-making approaches, such as experience-based versus. structured analysis,) are most effective within specific industries.

5. Facilitator Expertise and Workshop Design Gaps

- Incomplete risk identification frequently results from weak facilitation, low stakeholder participation, and inadequate communication. However, research does not examine how these factors vary across industries or experience levels.
- Current frameworks offer limited guidance for designing early-stage risk workshops that consider industry-specific thinking patterns and stakeholder backgrounds.
- Process standardization, time constraints, and management support all influence early risk identification, yet their interaction with industry context remains unclear.

6. Missing Evidence on Cross Industry Risk Perspective Integration

- When project managers transition between industries or join cross-functional teams, the literature provides no systematic data on which specific risks become unrecognized due to differing industry mental models.
- No studies quantify the cost or frequency of cross-industry misunderstandings during early-stage planning decisions. The practical question "How should a construction PM integrate IT risk perspectives when joining a software project?" lacks empirical answers.

7. Limited Research on Tools and Techniques Fit by Industry

- Although numerous risk tools exist, such as risk registers, checklists, matrices, and software, comparative studies rarely assess how project managers perceive tool effectiveness across industries or during early-stage planning.

- There are no evidence-based recommendations regarding which tools are most effective for early planning in construction, IT, or manufacturing, despite differing risk environments.

8. Insufficient Data on Management Pressure and Time Constraints

- Research acknowledges that time pressure and stakeholder expectations lead managers to underestimate early risks, but it does not examine whether this cognitive effect varies across industries.
- The influence of urgency on decision-making trade-offs in industry-specific early planning remains unmeasured.

1.6 Research Questions

The research questions are:

1. How do project managers in construction, IT, and manufacturing differ in their early-stage risk identification?
2. What decision-making patterns emerge when project managers face uncertainty in different industry contexts?
3. Which risk management tools and techniques work best in specific industries?
4. How can understanding these differences improve cross-industry project success rates?

1.7 Research gap

Despite a substantial body of literature on project risk management in general, several important aspects of early-stage risk work remain insufficiently explored. Based on the review of recent studies on risk identification, uncertainty, and project decision-making, six specific gaps can be identified.

1. Limited understanding of early-stage risk identification mechanisms

Many studies describe tools and processes for risk management, but they focus on later project phases or on generic process models. There is still little empirical insight into how project managers actually identify and frame risks in the very first 20% of a project, when information is incomplete and plans are still flexible. Existing work also indicates that early risk identification often fails to capture the risks that later turn out to be most critical for project performance.

2. Bias towards internal, familiar risks and neglect of external threats

Prior research shows that the majority of identified risks tend to be internal to the project and under the control of the project team, while external risks (e.g., political, regulatory, market or stakeholder-related) receive less attention, even though they can have severe effects. This suggests a systematic bias towards familiar and controllable risks in early risk identification. What is not yet well understood is how project managers could overcome this bias in practice and what helps them recognise low-likelihood, high-impact external risks at an early stage.

3. Insufficient exploration of cross-industry differences in risk perception

Most empirical studies are confined to a single sector, such as construction or IT, and therefore provide only a partial view of risk perception. Some comparative work indicates that risk is perceived and managed differently across industries and cultural contexts, but the underlying reasons for these differences have not been examined in depth. It remains unclear how a project manager's industry background shapes which risks come to mind first and which are systematically underestimated or ignored.

4. Limited insight into decision-making under uncertainty in the early phases

The literature acknowledges that project managers often have to make important commitments while key information is still missing, ambiguous or highly uncertain. However, there is relatively little empirical research on how they actually make such decisions in practice at the beginning of a project. Existing work points to the challenge of dealing with inadequate project information and the lack of a single, widely accepted framework for managing uncertainty in highly novel or complex projects, but it does not explain how project managers navigate these conditions in their day-to-day decision-making.

5. Under-researched industry-specific blind spots and transfer issues

There is growing recognition that each industry has its own "typical" risks and blind spots, yet systematic documentation of these patterns is still scarce. Very few studies investigate what happens when project managers move between industries, which risk habits transfer well and where previous experience might create dangerous assumptions. As a result, there is limited guidance for practitioners on what they need to unlearn or adapt when entering a new sector.

6. Limited understanding of how intuition and structured methods are combined in practice

Textbooks and standards describe a range of structured methods for risk assessment, but empirical studies suggest that project managers often rely heavily on their own experience and intuition and sometimes find formal techniques of limited practical use. What remains unclear is how experienced project managers actually balance intuition, informal judgement, and formal tools in the early stages of projects, and under which conditions each approach is perceived as helpful or hindering. This tension between prescribed methods and real-life practice has not yet been sufficiently explored.

1.8 Problem statement

Although project risk management literature extensively documents formal processes, tools, and risk categories, it offers limited insight into how project managers identify and respond to risks during the critical early planning phase, defined as the first 20% of a project, when information is mostly incomplete, and decisions have the greatest impact. Empirical studies indicate that many project failures and overruns originate from risks overlooked at this stage, despite the existence of established frameworks. However, these studies do not sufficiently explain the practical mechanisms, such as intuition, industry experience, or contextual pressures, that drive early risk awareness and prioritization across sectors, including construction, information technology, and manufacturing. This gap leaves practitioners without clear guidance for addressing industry-specific blind spots or transferring effective practices between contexts, thereby hindering improvements in early-stage planning and overall project success.

1.9 Aim and objectives of the study

This study investigates how experienced project managers in construction, information technology, and manufacturing identify risks and make decisions during the early planning phase, defined as the first 20% of a project, when information is most uncertain and incomplete. Using semi-structured interviews, the research seeks to uncover the practical mechanisms, including intuition, prior experience, formal tools, and contextual pressures, that influence early risk awareness and prioritization.

Research Objectives

1. Document industry-specific risk perception patterns by identifying the initial risks considered by project managers in various sectors and highlighting common blind spots that may be overlooked by those outside the industry.
2. Examine early-stage risk identification practices by analyzing the methods, such as brainstorming, checklists, and stakeholder input, that project managers employ during the first 20% of projects, and assess their suitability within specific industry contexts.
3. Analyze decision-making under uncertainty by investigating how project managers prioritize ambiguous factors and make decisions when critical information is lacking, with particular attention to the influence of time constraints and management expectations.
4. Compare cross-industry differences by revealing how risk awareness and decision-making habits vary between industries, and identifying challenges that arise when practices are transferred from one sector to another.
5. Derive practical implications by developing recommendations for training and tools that support improved early risk identification, particularly for project managers transitioning between industries or undertaking novel projects.

2 Significance of the Study

This study is important for both project management theory and real-life project work because it looks at how project managers in different fields find risks and make decisions in the first 20% of a project, when there is the most uncertainty and many problems arise later on. Previous studies have shown that identifying risks early on typically misses the most important ones, and that making decisions when you're not sure what to do is still not well understood in real projects. The study examines actual behaviours, perceptions, and blind spots across several industries, focusing on a relatively overlooked phase of the project life cycle and an incomplete body of knowledge regarding risk perception and early-stage decision-making.

2.1 Theoretical contribution

This study advances project management theory in several respects. It enhances understanding of early-stage risk identification by detailing how project managers recognize risks during the initial 20% of project timelines, moving beyond formal process models that typically assume rational and comprehensive identification. Through cross-industry narrative comparisons, the research offers empirical insights into which risks are prioritized, which are overlooked, and how practitioners balance internal and external risks in real-world settings.

Additionally, the study extends understanding of risk perception and decision-making under uncertainty by analyzing how project managers alternate between intuition, experience, and formal tools when faced with incomplete information. While previous research has acknowledged the difficulties posed by insufficient information and the limitations of current frameworks for managing uncertainty, few studies have demonstrated how managers navigate these challenges at the project outset. The findings provide a nuanced perspective on how contextual factors such as industry, project type, and organizational pressures influence risk-related judgments, thereby contributing to theoretical discussions on project uncertainty, human judgment, and contextual influences in risk management.

Furthermore, by incorporating project managers from multiple industries, the study contributes to cross-industry comparisons of risk management practices. Although empirical research has highlighted cultural and contextual variations in risk perception, it has seldom systematically linked these differences to early-stage planning behaviours and industry-specific blind spots. This thesis addresses this gap by identifying both common and context-dependent patterns, thereby informing future theoretical development regarding the influence of domain experience and organizational environment on risk awareness.

2.2 Practical Contribution

In practice, the study gives organizations and project managers ideas on how to better deal with risks at the outset of projects. By writing down the usual early hazards, the best ways to find them, and the common blind spots in different fields, the results can assist project managers compare their own patterns to those of their peers and find risks they might be missing. This is especially useful because many projects that go over budget or fail have been connected to hazards that weren't spotted early enough, even when there are formal risk

management procedures in place.

The results also give useful information for training, onboarding, and moving between industries. As project managers switch from one field to another (for example, from IT to construction or manufacturing), they bring with them certain risk habits that might not work in the new setting. This could lead to misunderstandings or blind spots. The study can help project managers who are moving to a new industry by showing them which methods work well and which assumptions are risky. The focus will be on early-stage planning and risk awareness.

Lastly, the study can help businesses improve their tools and methods for managing risk. The research shows where tools are seen as useful, where they are ignored, and why by comparing formal approaches with how project managers really utilize (or don't use) them in the early phase. This can help make risk registers, checklists, and early planning routines better by making them more in line with how people actually make decisions. This will help identify risks earlier and make projects more likely to succeed in a variety of fields.

3 Literature Review

The field of Project Risk Management (PRM) has usually been characterized by a "task perspective," emphasizing the technical detection, evaluation, and alleviation of objective threats to project restrictions such as time, cost, and quality. The persistent occurrence of project failures, particularly those shown as cost overruns and schedule delays, is evidence that merely adhering to standardized procedures is not sufficient. The behavioural and cognitive features of the project manager (PM) have received an increasing amount of attention in recent and recent study. It is becoming widely acknowledged that risk is not merely a mathematical chance but rather a subjective reality that is mediated through the cognitive filters of people who are engaged in early-stage planning.

In this literature review, modern research on risk awareness, cognitive biases, and decision-making processes are compiled and analyzed. In addition, it investigates the ways in which these factors differ across other professional domains, specifically comparing and contrasting the Engineering and Construction, Information Technology (IT), and Manufacturing industries. The purpose of this review is to uncover gaps in the understanding of early-stage planning across a variety of situations by defining the intersections between human cognition and industry-specific limitations.

3.1 Synthesis Table

A comprehensive literature review was conducted to establish the theoretical and research foundation of this study. A total of ten scholarly papers from diverse sources were thoroughly examined to understand existing knowledge, gaps, and methodologies.

Table 1: Synthesis of Literature Review

Title of the Paper / Work	Author	Aim	Observation/ Understanding
Project Managers and Decision Making: Conditional Cognitive Switching and Rationally Stepping Up	Nowińska & Pedersen (2024)	To investigate how project managers toggle between intuitive and analytical thinking when facing high uncertainty.	The authors observe that effective project managers are "conditional cognitive switchers." Unlike the assumption that PMs should always be analytical, this study finds they rely on intuition ("System 1") for routine tasks but "rationally step up" to analytical ("System 2") modes specifically when project uncertainty increases. This supports the "Cognitive Pillar" by

			showing that expertise is the ability to switch modes, not just follow a process.
The Effectiveness of Risk Management: An Analysis of Project Risk Planning Across Industries and Countries	Zwikael & Ahn (2011)	To analyze the variation in risk management maturity and planning effectiveness across different industries and national cultures.	The study proves that "one size does not fit all." Construction and Engineering show high risk maturity due to tangible constraints, while other sectors vary significantly. Crucially, the authors found that risk planning has a direct correlation with project success, but this relationship is moderated by the industry context and the cultural background (e.g., Uncertainty Avoidance) of the project team.
Balancing Expertise And Experience: A Comparative Study Of Risk Management Practices	Soler Del Toro (2023)	To compare the risk management behaviours of certified (e.g., PMP) versus non-certified project managers.	The research reveals a distinct behavioural gap: certified managers rely heavily on structured methodologies and formal tools (the "Process Guardian" archetype), whereas non-certified managers lean on intuition, adaptability, and "tacit knowledge" gained from experience. This supports the argument that certification may drive "Defensive Decision Making" by prioritizing compliance over agility.
Risk Perception-Based Project Contingency Management Framework	Ottaviani et al. (2024)	To provide a mathematical framework for managing financial buffers based on the project manager's perception of risk.	The authors argue that contingency management is often subjective. They distinguish between "Short-Term Concern" (immediate budget deviations) and "Long-Term Concern" (forecasted variance). The study highlights

			that a PM's perception of these concerns dictates whether they release funds to mitigate risk or "hoard" budget, often driven by a fear of running out of resources rather than objective data.
Biases in Project Estimating	Glauser (2025)	To identify the specific cognitive biases that lead to consistent underestimation of project time and costs.	This work anchors the "Economic Pillar" by identifying "Optimism Bias" (believing the future will be better than the past) and "Anchoring Bias" (sticking to the first number heard) as primary drivers of project failure. It suggests that without intervention, human nature drives PMs to underestimate risk severity during the early planning stages.
Prospect Theory: An Analysis of Decision under Risk	Kahneman & Tversky (1979)	To critique Expected Utility Theory and introduce a model of decision-making under risk based on psychological realism.	This foundational text establishes the "Economic Pillar" of your thesis. It posits "Loss Aversion"—the observation that the pain of a loss is psychologically more powerful than the pleasure of an equivalent gain. In project management, this explains why PMs often pay a "safety premium" or choose sub-optimal paths to avoid the perception of a loss (e.g., a budget overrun).
Sources of Power / Naturalistic Decision Making	Klein (1998/2008)	To understand how experts make high-stakes decisions in time-pressured, uncertain environments.	This supports the "Cognitive Pillar" by introducing "Recognition-Primed Decision" (RPD) making. Klein observes that experts do not compare options (as standard PM tools suggest);

			instead, they use "Pattern Recognition" to match the current situation to past experiences. This validates the finding that veteran PMs rely on "Somatic Verification" (gut feel) over digital dashboards.
Risk Savvy: How to Make Good Decisions	Gigerenzer (2014)	To explore how heuristics and simple rules of thumb can be superior to complex calculations in uncertain worlds.	This resource anchors the "Defensive Pillar." Gigerenzer identifies "Defensive Decision Making," where professionals choose options that protect them from blame (CYA-Cover Your Ass) rather than the best option for the project. This aligns with the finding that high-liability roles use tools primarily as an "Audit Shield" rather than for actual risk mitigation.
Culture's Consequences	Hofstede (2001)	To map the influence of national culture on organizational behaviour and values.	This is the backbone of the "Cultural Pillar." Hofstede's dimension of "Uncertainty Avoidance" provides the theoretical basis for why German PMs ("Process Guardians") may insist on rigid adherence to plans to mitigate anxiety about the future, whereas cultures with lower uncertainty avoidance may be more comfortable with ambiguity.
Jugaad Innovation	Radjou, Prabhu, & Ahuja (2012)	To describe a flexible, frugal approach to innovation and problem-solving common in emerging markets.	This concept supports the "Resource Navigator" archetype found in your research (specifically regarding Indian management styles). It frames the ability to improvise solutions with limited resources not as a lack of

			process, but as a sophisticated cultural capability for managing high uncertainty without heavy "safety premiums."

3.2 Findings on the "First 20%" and the Human Element in Early Planning

Recent research indicates that the initial phase of a project, often referred to as the "First 20%" of its lifecycle, is the most critical period for managerial influence (Arge et al., 2004, pp. 545-551). Edkins et al. (2013) characterize this "front-end" as offering the greatest opportunity to add value, while also presenting significant risk, as decisions made with limited information can have enduring negative consequences. In sectors such as construction and manufacturing, this stage is defined by Physical Irreversibility. Unlike software, which can be modified or deleted, materials like concrete and steel cannot be easily reversed once utilized (Compressive behaviour deterioration mechanism of concrete under carbonization and freeze-thaw cycles and time-varying structural reliability evaluation, 2025).

As a result, the substantial cost associated with early-stage errors contributes to a "Trust Deficit" between project managers (PMs) and digital tools. The literature indicates that experienced managers often distrust automated dashboards and quantitative reports, frequently regarding them as "Polite Fictions" that obscure the complex realities of the job site (Beebe, 1995, pp. 153-168). Rather than relying exclusively on digital artifacts, PMs tend to exhibit a "Universal PM DNA," a set of cognitive behaviours identified in the literature that emphasizes psychological safety and intuitive verification over reliance on raw data (Santana et al., 2025).

This human dimension of project management is manifested in three primary ways:

• Loss Aversion: According to Prospect Theory, managers are highly motivated to avoid losses such as schedule delays, even if this requires incurring additional costs that may appear excessive (Kahneman & Tversky, 1979; Glauser, 2025). Consequently, they frequently pay a "safety premium" to mitigate uncertainty.

• Defensive Decision Making: Research demonstrates that managers frequently employ formal tools primarily to shield themselves from blame rather than to enhance project outcomes. This aligns with Gigerenzer’s (2014) concept of defensive decision-making, in which the avoidance of responsibility takes precedence over achieving optimal results.

• Somatic Verification: In high-pressure contexts, experienced managers utilize Naturalistic Decision Making (NDM). Rather than solely analyzing available options, they engage in "Somatic Verification" by directly observing, listening, or physically inspecting the site to gain situational awareness (Klein, 2008). These managers place greater trust in their own sensory assessments than in digital dashboards for identifying problems and risks.

Ultimately, research demonstrates that early project planning is both a technical and cognitive process. Project managers exercise personal judgment to assess risks, frequently depending on intuition and defensive strategies rather than exclusively adhering to digital tools (Baumgartner, 2016).

3.3 The Economics of Certainty: Why Managers Pay a Safety Premium

To understand why project managers overspend early on, we have to look at Prospect Theory. Kahneman and Tversky (1979) identified a key behaviour called Loss Aversion basically, human beings hate losing about twice as much as they enjoy winning.

3.3.1 The Psychology of Loss Aversion

Understanding why project managers frequently overspend during early planning stages requires an examination of Prospect Theory. Introduced by Kahneman and Tversky (1979), this framework challenges traditional economic theory by presenting the concept of Loss Aversion. Loss Aversion refers to the finding that the psychological impact of a loss is significantly more powerful than the satisfaction derived from an equivalent gain.

Within project management, this asymmetry suggests that managers experience greater psychological distress from a two-day schedule delay than satisfaction from completing a project two days early. This dynamic contributes to the Certainty Effect, in which decision-makers strongly prefer guaranteed outcomes over probabilistic alternatives. As a result, managers may be willing to incur a disproportionate financial cost, effectively paying a "safety premium," to eliminate risk entirely rather than choose a less expensive but uncertain option. This behaviour prioritizes the avoidance of a "hard stop" or total project failure over the pursuit of optimal resource efficiency.

3.3.2 The Safety Premium as a Survival Reflex

This study introduces the term "Safety Premium" to describe how project managers often allocate extra resources early in a project. The literature calls this a "Strategic Redundancy Reflex," where managers use financial buffers and order more materials than needed, not just for technical reasons but also to protect against uncertainty.

Research shows that managers often see the budget as something they can refill, but view time and reputation as limited and impossible to replace. (G. & G., 2020, pp. 5-16) By spending extra early on, they are buying certainty. This matches the Certainty Effect from Prospect Theory, where people pay more to avoid the risk of a project stopping completely. (The effect of decision risk and project stage on escalation of commitment, 2007, pp. 225-237) Managers would rather explain a reasonable budget overrun than face the reputational damage of a stalled project.

3.4 Expertise and the Human Sensor: The Value of Somatic Truth

3.4.1 Naturalistic Decision Making (NDM)

In fast-paced fields like construction, managers rarely get to weigh options in a controlled way. Instead, they rely on Naturalistic Decision Making (NDM). According to Klein (2008), experts do not "calculate" decisions; they recognize them. By spotting patterns, experienced managers unconsciously match the current situation to memories of many past experiences. This helps them spot the type of problem and notice risks before computerized systems do.

3.4.2 Somatic Verification vs. Polite Fictions

This dependence on experience leads to what is called Somatic Verification. Research shows that experienced managers often do not trust digital dashboards, seeing them as "Polite Fictions" or cleaned-up reports that leave out the messy "noise" from the job site. For experts, that noise is important information. Managers would rather "walk the site" or "listen to the machine" because their senses can pick up on things like a strange vibration or a change in work pace that software cannot detect. For them, what they see and feel on site is the most reliable source of truth.

3.5 The Defensive Manager: The Role of the Bureaucratic Shield

3.5.1 Defensive Decision Making (The "CYA" Culture)

The last section explained how managers use their senses to find the truth. This section looks at why they sometimes use paperwork to avoid blame. This is called **Defensive Decision Making**. Gigerenzer (2014) says this happens when a professional picks an option that protects them from liability (a "Cover Your Ass" or CYA strategy) instead of the one that would give the best result for the project.

In organizations where blame is common, managers focus more on Defensibility (showing they followed the rules) than on **Optimality** (being efficient). Research suggests this defensive approach often leads to less effective decisions, where the main goal is personal survival in the organization, not project success.

3.5.2 The Risk Register as an Audit Shield

This study describes something called the "Bureaucratic Shield." The findings show that jobs with high risk, like working with medical devices or tunnel construction, rely heavily on formal paperwork. Managers in these roles often see the Risk Register not as a way to manage risks, but as an "Audit Shield" a tool to shift responsibility.

The thinking is simple: "if it isn't documented, you are personally liable." Because of this, managers create a lot of paperwork to protect themselves. This matches Soler Del Toro's (2023) findings that certified managers often depend on formal tools as a "scaffold" for making

decisions, which can end up putting compliance ahead of flexibility. Here, the risk register is used less as a guide and more as protection against being blamed later.

3.6 The Physics of the Medium: Irreversibility and Physical Debt

3.6.1 Path Dependency and the One-Shot Mindset

Path Dependency limits managers in the field of physical projects. In digital contexts, code may be changed, but in physical construction, once concrete is poured or steel is cut, the choice is permanent. Project managers develop a One-Shot Mindset because they know that decisions made early on define a path that is very hard to change later on.

This situation brings up the idea of Physical Debt. IT projects typically include technical debt that can be refactored (corrected) in future updates. Physical mistakes, on the other hand, are irreversible and cannot be paid off. Because of this, managers have to put fact-based complexity dealing with interconnected physical information above the flexibility that is commonly seen in software development.

3.6.2 The Rejection of Agile in Heavy Industry

As a result, the results show that heavy industry is quite resistant to Agile methods. For example, 87% of managers said that iterative methods were reckless for physical assets. Construction managers think that Agile ignores the enormous cost of making mistakes with physical things, while it works well for learning and making decisions that are flexible and not centralized.

This makes people willing to pay a Safety Premium. Managers do not see over-resourcing as a waste; they see it as a required buy of certainty against Commitment Bias, which is the desire to stick with a poor course of action since you cannot get back the money you put in. In this case, the Safety Premium buys protection against the Undo Button, which is not real in the real world.

3.7 The Impact of Culture: The Guardian vs. The Navigator

3.7.1 The German Process Guardian: A Defense Against the Law

This study indicates a unique archetype among German managers: the Process Guardian. Hofstede defined Uncertainty Avoidance as the degree to which a society feels threatened by unclear situations. This tendency is strongly ingrained in that.

For these managers, the biggest danger is typically not physical failure, but Regulatory Paralysis, which means the risk of not following the law. As a result, they see strictly following digital tools and rules as a smart way to stay alive. The instrument acts as a legal shield when the "Perfect Plan" is strictly followed. If the project fails but the process was followed, the manager is not personally responsible.

3.7.2 The Indian Resource Navigator: A Defense Against Chaos

In India, on the other hand, most managers are Resource Navigators. They can't depend on strict plans since they work in an environment where things might change quickly, such when supply chains break down or the weather gets really bad. Instead, they use Contextual Adaptability, which means they change their plans on the fly to fit the changing situation on the site.

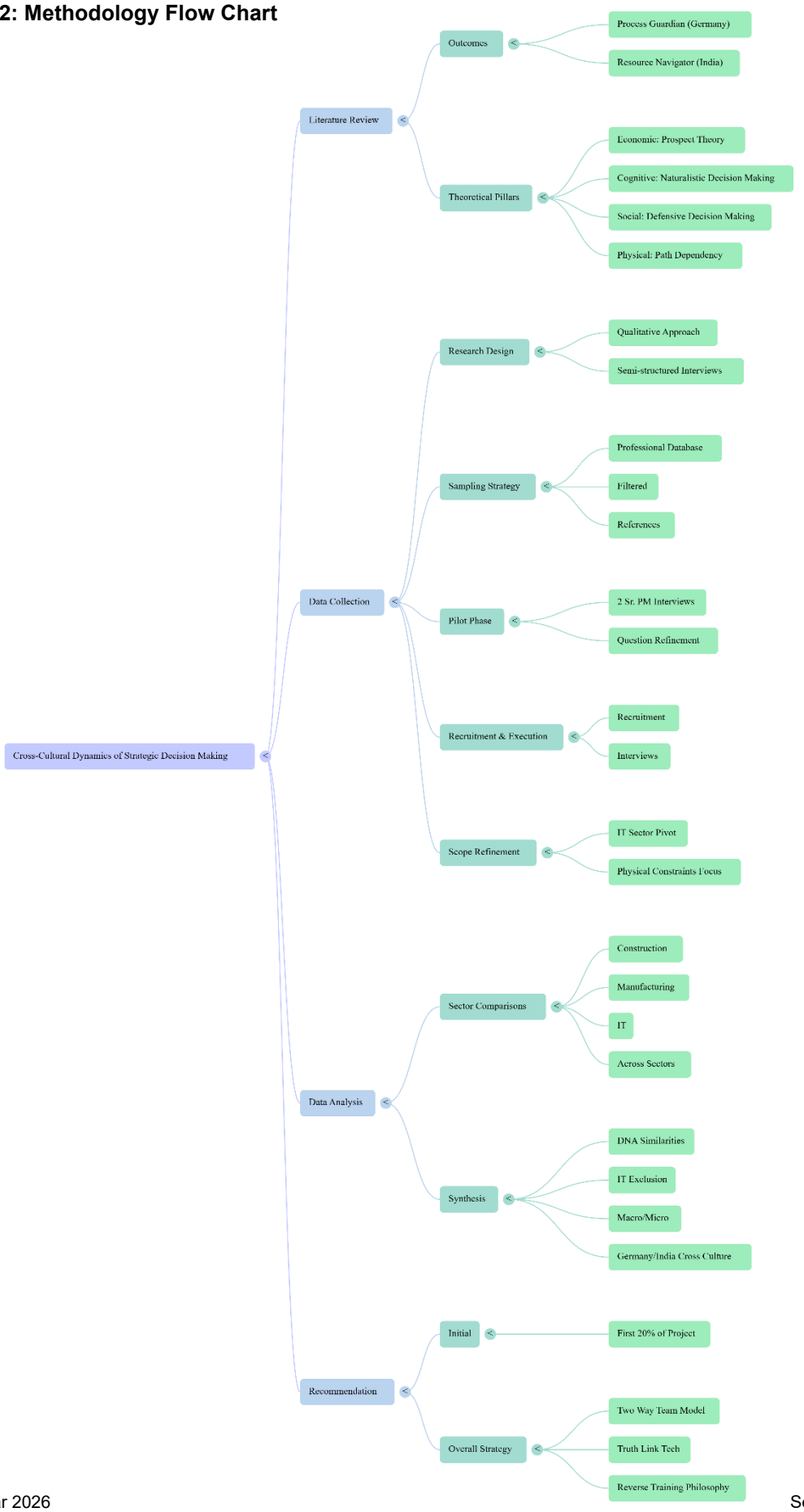
This method is in line with the idea of Jugaad, which is a cultural practice of being flexible and creative with few resources to address challenges. The Navigator sees a digital report as just counsel, and the only single source of truth they trust to make judgments is the actual site.

4 Methodology

This chapter elucidates the methodical methodology employed to examine how project managers discern risks and make decisions at the pivotal phase of a project the initial planning stage. A lot of research on project management looks at the technical or administrative parts of it. This study, on the other hand, looks at the people involved. It looks at the cognitive reflexes and gut feelings that leaders have when they don't have all the knowledge they need and the pressure is intense. This study focuses on the initial 20% of a project's lifecycle, as this period signifies a dynamic moment during which project managers exert the greatest impact on the ultimate result, while having the least quantity of verified information. The research seeks to identify the Universal PM DNA by examining the actual experiences of professionals in the construction, manufacturing, and information technology sectors, focusing on the common behaviours that enable managers to connect digital plans with tangible realities.

The study examines two very distinct industrial and cultural settings in order to give a full picture of the world. The study examines Germany, noted for its stringent regulatory oversight, emphasis on procedural inflexibility, and significant dependence on digital tools as a professional safeguard. In contrast, India is a fast-paced and often unpredictable place where managers have to be resourceful and use their social and technical skills to keep things moving forward. This chapter uses a qualitative technique that focuses on in-depth anecdotes and thorough interviews to show how the lived reality of project management was recorded in these different cultures and industries.

Figure 2: Methodology Flow Chart



4.1 Literature Review

The Literature Review functions as the theoretical foundation for this study, offering the essential framework for interpreting the real-world experiences of project managers. Before analyzing the interviews, it was important to look at how experts and past researchers have characterized risk and decision-making in high-stakes situations. This was necessary to construct a strong basis. This aspect of the technique isn't just about gathering historical data; it's also about figuring out the best ways to explain why managers do what they do when things become tough. The study looks at current studies to make sure that the final results are not just random opinions but are also part of bigger scientific discussions on how industries work and how people think when they are under pressure.

The review concentrates on three principal domains essential for comprehending project success during the initial phases. First, it looks at the history of risk management to see how standard techniques and checklists have been used to try to bring order to uncertainty. Second, it looks at elements that are distinctive to each business and compares how diverse settings, like a big construction site, a high-tech production floor, or a virtual IT office, put different kinds of stress on the individuals in charge. Lastly, it looks at the human side of things, such as how a manager's years of experience and cultural background might affect whether a danger is noticed early or missed altogether. This thorough backdrop enables the research to progress with a lucid comprehension of the existing state of project management prior to the introduction of new models for enhancement.

4.2 Data Collection

The main goal of the data collecting phase was to record the lived experience and thought processes of Project Managers (PMs) during the first 20% of a project's life cycle, which was quite unstable. A qualitative strategy was selected to attain this level of understanding, focusing on semi-structured interviews with field practitioners. This approach facilitated an in-depth examination of the ways in which managers assess risks and make decisions in uncertain situations, concentrating on the physically irreversible contexts of Construction and Manufacturing within two unique cultural settings: Germany and India.

4.2.1 Research Design

A qualitative research design was employed to thoroughly examine the intricate, experience-driven decision-making processes of project managers. This methodology was used instead of quantitative methods since the study sought to investigate the fundamental "how" and "why" of human behaviour, rather than to quantify or tally predetermined characteristics. The objective was to produce comprehensive, detailed data regarding perceptions, cognitive processes, and implicit knowledge that are challenging to obtain solely through surveys.

In this qualitative approach, the principal instrument for data collection was the semi-structured interview. This format struck the right mix between being consistent and being able to change. All participants were led through the same main ideas and research questions, which made it possible to compare the interviews in a reliable way. At the same time, the semi-structured

format permitted the interviewer ask more questions about fascinating, unexpected answers, follow different lines of inquiry, and let participants talk more about their own experiences and tales. This strategy was crucial for revealing the "lived experience" and contextual rationale that characterize risk awareness in practice.

4.2.2 Sampling Strategy

A targeted, multi-stage sampling strategy was used to recruit participants with direct and relevant experience in the key sectors and regions for this study. This approach aimed to provide deeper insights rather than broad statistical coverage, which fits the qualitative focus of the research.

The participant pool came from a professional network of about 200 to 300 Project Managers, built during the researcher's previous industry work. This made it possible to reach experienced professionals who are often hard to contact through cold outreach. From this group, participants were chosen based on three clear criteria: their main industry sector (Construction, Manufacturing, or IT) and their country of operation (Germany, elsewhere in Europe, or India). To add more variety, especially by including junior-level Project Managers, a referral or snowball sampling method was also used. Early participants suggested colleagues who fit the study's criteria, helping to build a trusted network of contacts.

4.2.3 Interview Question Preparation

I created the semi-structured interview protocol to explore the main research problem: the gap between formal risk management processes and how decisions are actually made under uncertainty. My goal was not just to list the tools managers use, but to understand why and when they use them, or choose intuition instead.

I organized the questions to move naturally from what managers do to why they do it. I started with straightforward questions about current projects and usual procedures to set the context and build trust. Then, I asked more reflective questions to reveal hidden factors like concerns about liability, the role of seniority, and how people respond to uncertainty or pressure from management. This step-by-step approach helped uncover the unspoken knowledge and instincts that shape real decisions.

4.2.4 Pilot Phase

Before collecting the main data, a pilot phase was carried out to check and improve the interview questions.

1. Two Senior Project Managers Interview

The researcher interviewed two experienced Project Managers from their professional network using the first draft of the questions. The main goal was to see if the questions were clear, relevant, and made sense in practice. These senior managers gave helpful feedback on the wording and how useful the questions would be in real situations.

2. Interview Question Refinement

After getting feedback, the interview protocol was updated. The main changes focused on making the Management Pressure questions clearer so participants could give specific examples about balancing schedule demands and risk tolerance. This step-by-step process helped make sure the final questions were clear, useful, and matched the research goals.

4.2.5 Recruitment

I used a personal and direct approach for recruitment, relying on the professional relationships I had built as a project coordinator. First, I contacted Project Managers I already knew, since our trust made them more open to sharing their experiences. I then asked these contacts to refer me to other qualified peers in their network, such as junior colleagues or specialists in certain sectors, since their introductions would help create a warmer connection. This referral method helped me expand my participant pool through trusted channels. After getting contact details, I reached out to each referral directly to explain the study and arrange an interview.

4.2.6 Data Collection

I collected the main data for this study through a series of semi-structured interviews, using the protocol and sampling strategy described earlier. During this phase, I focused on carrying out the interviews smoothly and making sure each participant had a consistent experience.

I held all interviews online from late November 2025 to late January 2026 to accommodate participants in different countries. Each session lasted about 30 minutes, which balanced the Project Managers' busy schedules with the need for in-depth discussion. To keep things consistent for both the German and Indian groups, I conducted every interview in English. I used the updated semi-structured protocol to cover important topics, but I also stayed flexible to explore unique insights from each participant.

4.2.7 Scope Refinement

At the start of data collection, I made an important change to the study’s scope to improve the comparative analysis.

The original research plan included the Information Technology (IT) sector as a third industry for comparison. After the first interviews with IT Project Managers, I noticed that the types of risk they faced were quite different. Their main concerns were about logical dependencies, code integration, and virtual scalability, not the physical and material limits that are key in Construction and Manufacturing. To keep the study focused on 'physical irreversibility,' I decided to change direction. I set aside the IT data to use only as background, and focused the main analysis on Construction and Manufacturing, since both deal with real physical constraints.

4.3 Data Analysis

I analyzed the interview data with the study's comparative goal in mind. I worked to identify, compare, and contrast the main patterns, themes, and decision-making approaches described by the Project Managers.

I started by reviewing the notes and transcripts from each interview, paying close attention to how participants described their experiences, attitudes, and strategies. My goal was to find the main insights that explained their behaviour. I looked for repeated ideas within each industry group (Construction and Manufacturing) and each cultural group (Germany and India), and then compared the differences between these groups.

4.3.1 Construction Sector Analysis

Interview data from Construction Project Managers was divided into focused themes. This made it easier to examine their views on risk, how they identify risks, and how they make decisions, as described in the findings. The analysis followed the order of a project's early stages, from first looking at risks to final outcomes.

4.3.2 Manufacturing Sector Analysis

Interview data from Manufacturing Project Managers was analyzed using the same approach. This helped clearly show their unique risk landscape, which is shaped by supply chain issues and regulatory pressures, as well as their ways of identifying risks and making decisions, as described in the findings. The analysis also followed the order of a project's early stages, from first looking at risks to practical outcomes.

4.3.3 IT Sector Analysis

Interview data from IT Project Managers was also analyzed using the same method. However, IT risks, which focus on logical dependencies, integration issues, and virtual scalability, are very different from the physical risks in Construction and Manufacturing. For this reason, the IT data was set aside as a comparison point to highlight the unique challenges of virtual projects and was not included in the main cross-industry comparison.

4.3.4 Cross-Sector Analysis

After analyzing each sector, a comparative analysis was done across all three industries. This process involved systematically comparing the themes, behaviours, and decision-making approaches from Construction, Manufacturing, and IT. The goal was to identify both sector-specific patterns and universal behaviours, which led to the frameworks presented in the findings, such as the horizontal analysis segments and the core elements of the "Universal PM DNA."

4.3.5 Universal Similarities (The PM DNA)

The cross-sector analysis found key behaviours shared across industries, called the "Universal PM DNA." Although the risks differ, managers in Construction, Manufacturing, and IT showed similar psychological habits. These include the Strategic Redundancy Reflex (90% prefer overspending for certainty), Methodological Scepticism (93.3% are wary of formal digital reports), and the Optics Bias (focusing on visible progress under pressure). This suggests that core project management behaviours are shaped more by common human psychology in uncertain situations than by industry tools.

4.3.6 Exclusion Rationale

A decisive step in the analysis was the methodological exclusion of the IT sector from the final comparative framework. This decision was made to preserve the internal validity and clarity of the study. The analysis revealed that the IT sector's internal variance in philosophies and practices such as the conflict between Agile experimentation and regulatory stability was too great to treat it as a single, comparable industry. More fundamentally, the core concept of reversibility creates an unbridgeable divide: IT's world of "virtual refactoring" and patching operates on fundamentally different psychological and material principles than the "physical irreversibility" that defines Construction and Manufacturing. This exclusion sharpened the focus onto the stable, high-stakes environment of physical project management, governed by unyielding material constraints.

4.3.7 Macro/Micro Experience Analysis

Within the group of Construction and Manufacturing managers, a special analysis examined how professional experience affects risk perception. The study compared junior managers (1-5 years of experience) with senior managers (over 20 years). It aimed to compare their main risk concerns, whether they focus on technical details (micro-level) or on strategy and legal issues (macro-level), and to see how these experience levels shape decisions and possible blind spots in planning.

4.3.8 Cross-Cultural Archetype Analysis

A dedicated analysis was performed to compare the risk management approaches between the German and Indian participants. This examination identified two dominant, culturally-informed archetypes: the German "Process Guardian," characterized by a shield mentality and high reliance on digital tools for compliance, and the Indian "Resource Navigator," defined by a suggestion mentality and adaptive, often improvisational, responses to environmental volatility. The analysis focused on contrasting their fundamental approaches to tool usage, environmental adaptation, and their respective professional blind spots.

4.4 Recommendations

Based on the systematic analysis of the interview data, this section translates the key findings into structured, practical recommendations. These proposals are designed to address the specific gaps and behavioural patterns identified in the study, offering actionable strategies for organizations to improve risk awareness and decision-making in the critical early stages of physical projects. The recommendations focus on human-centric solutions, technological integration, and cultural adaptation.

4.4.1 Initial Phase

1.1.1.1 The 20% Reality Check

A structured Calibration Workshop is recommended at the 20% project completion mark. This intervention is designed to formally validate both technical and strategic project foundations before further commitment.

4.4.2 Overall

1.1.1.2 Two-Way Team Model

The Collaborative Risk Auditing Framework

This model sets up a two-way auditing process. Junior project managers handle technical micro-checks of on-site details, while senior managers guide them on legal and regulatory risks.

1.1.1.3 Truth-Link Technology

Sensor-Driven Verification

This method suggests using live physical sensors, such as those for concrete curing or machine acoustics, and connecting them to project dashboards to build a real-time Digital Twin. This helps solve the trust deficit by giving managers a steady flow of data that feels as reliable as checking things in person.

1.1.1.4 Reverse Training Philosophy

Bidirectional Cross-Cultural Training

This recommendation suggests a structured reverse mentorship program. In this setup, German project managers would learn adaptive problem-solving from their Indian colleagues, while Indian project managers would learn formal digital standardization from their German counterparts. The goal is to help both groups develop a mix of managerial skills.

5 Data Collection

5.1 Thought Process and Research Design

This phase of data collection was designed to break down the heuristics and tacit knowledge that Project Managers (PMs) use during the critical first 20% of a project. The study uses a qualitative, semi-structured interview approach instead of a quantitative survey. This choice fits the interpretivist nature of the research, as understanding how leaders handle extreme uncertainty means capturing the reasons behind their decisions. Such depth is often lost in the fixed categories of multiple-choice surveys.

The study focuses on the Construction and Manufacturing sectors instead of IT or Digital sectors to set a baseline of physical inelasticity. In these fields, changing course is harder because of material lead times, labour needs, and safety regulations. By looking at industries where mistakes cannot be undone, the research can better isolate the mental pressures of early planning. This approach removes the unpredictable and forgiving 'fail-fast' factors found in IT, making it easier to see how project foundations are actually built.

5.2 Sampling and Participant Recruitment

The selection process for this study employed a multi-stage, purposive sampling strategy. The objective was to surpass a generalist perspective by assembling a respondent pool with substantial, practical expertise in high-stakes environments. This methodology ensured that the insights obtained were grounded in the practical complexities of the manufacturing and construction sectors, rather than being solely theoretical.

5.2.1 The Sampling Frame and Reflexivity

The main group of participants came from a professional network of about 200 to 300 Project Managers. My experience as a Global Project Coordinator helped with this step. Because I had managed international logistics and worked with teams in Europe, Asia, and North America, I had the professional connections needed to avoid traditional barriers.

Choosing to use my insider status was intentional. Senior Project Managers in construction and manufacturing are often hard to reach through cold calls or outside contacts because of their demanding jobs. By using a database of trusted contacts, I was able to get more open and detailed interviews than an external researcher might have. My background also made it easier to use shared technical language with participants, which helped us explore early-stage project risks in more detail.

5.2.2 Recruitment Strategy

The "Trust-Based" Model used three specific sampling techniques:

- Purposive Sampling: I selected participants from the database based on their industry Construction, IT, or Manufacturing and their location in either Europe or India.
- Convenience Sampling: I first reached out to project managers I had worked with before. This helped build trust and encouraged more open responses about project failures and "gut feelings."
- Snowball Sampling (Referral Method): To fill gaps in the participant group, such as finding younger project managers in certain areas, I used referrals. I asked my main contacts to recommend colleagues or subordinates who fit the study's strict criteria, which helped keep the process accountable.

5.2.3 Outreach via LinkedIn

Although I already knew many of the participants, I used LinkedIn to reconnect and arrange interviews. I shared the research abstract there and explained the purpose of the interview. Using LinkedIn helped keep our communication professional and let participants see my academic background and the focus of my thesis.

5.3 The Interview Instrument: Design and Pilot

The interview instrument was developed in a step-by-step way, starting with objective, fact-based questions and moving toward deeper, more subjective psychological topics. This approach aimed to address the "theory-practice gap," which is the difference between formal project management tools and the practical, intuitive methods professionals use in high-pressure situations. By designing the questions to focus on this issue, the instrument aimed to find out how often professional intuition takes precedence over traditional risk-management frameworks during the early, uncertain stages of a project.

A specific pilot phase was completed within my existing professional organization to ensure the instrument's reliability and technical resonance. I conducted two first interviews with senior Project Managers, who assessed the question set for its practical applicability and clarity. This phase was crucial for ensuring that the academic terminology of the thesis was successfully conveyed in the participants' vernacular. The parts on "Management Pressure" were substantially enhanced based on the comments from leading specialists. The pilot participants observed that the initial questions were excessively broad to effectively address the unique tension between organizational demands for expediency and the operational requirement for risk mitigation. As a result, the final instrument was recalibrated to more accurately examine these trade-offs, so enhancing the "content validity" of the primary study.

5.4 Data Collection Procedure

Data collection took place over three months, from late November 2025 to late January 2026. Because participants were based in both Germany and India, all sessions were held online using platforms like Zoom and Google Meet. Using digital meetings made it possible to include senior perspectives from different time zones and gather a wide range of operational contexts, without the challenges of arranging travel.

Each interview lasted about 30 minutes on average. This time limit was chosen to respect the busy schedules of senior Project Managers. In manufacturing and construction, time is valuable, so keeping interviews short helped ensure that senior professionals could take part. Even with this shorter format, the semi-structured interviews allowed for in-depth discussion, focusing on the most important parts of early-stage planning instead of less relevant details.

All interviews were conducted in English to keep the language consistent for both German and Indian participants. This approach avoided issues with translation and made sure that the data could be analyzed in the same way for everyone, allowing for fair comparison across cultures and industries.

5.5 Recording and Data Integrity

To maintain data integrity and respect participants' professional boundaries, I used a hybrid approach to collecting data. This approach was designed for the Construction and Manufacturing sectors, where sharing operational failures or strategic mistakes can be risky. Most sessions were documented through detailed, real-time note-taking. This manual process was not just a backup for recording; it helped me identify key themes and highlight important quotes as they came up, without making participants feel observed as a recording might.

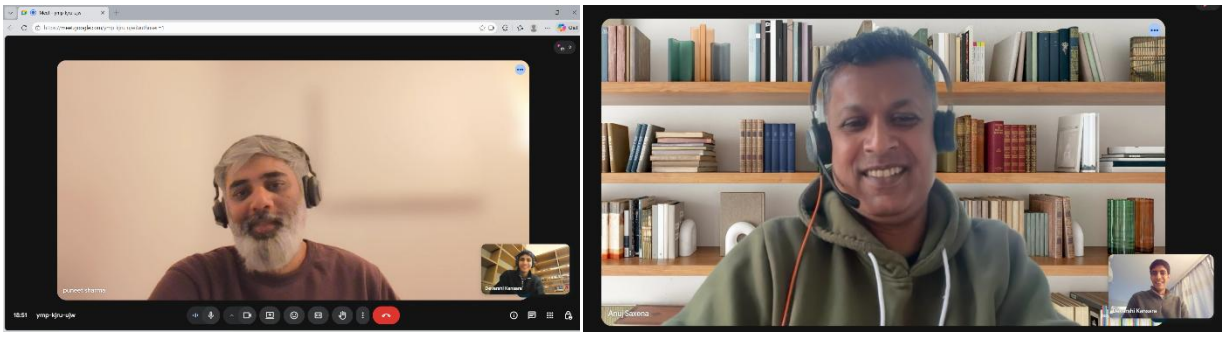


Figure 3: Example of online interview

When I already had a good working relationship with a participant, I recorded the interview after getting clear verbal consent. These recordings were especially useful because they provided more detail and allowed me to explore honest insights that might not come up in notes.

Using these relationships, I could better understand the psychological factors behind decision-making in the early stages of a project. This two-part method kept the data strong by combining the detailed information from recordings with the wider coverage of annotated interviews.

Because these industries are highly competitive, keeping participants' information confidential was essential. Some Project Managers shared sensitive details about project values and unique operational challenges. To protect their reputations and company secrets, I removed any identifying information. The final analysis focuses on the decision-making process and thinking patterns, not on specific projects. This way, the research could look at the reasons behind project management choices without risking anyone's business interests or ethical standards.

5.6 Evolution of Scope: The IT Sector Pivot

Good academic research means knowing when the original scope no longer fits the study's needs. Early in data collection, I decided to move our main focus away from IT and software development. Although my first plan was to look across different sectors, the first interviews showed a basic difference between digital and physical project environments. Risks in IT, such as issues with code or system design, do not have the same hard limits as those in the physical industries we studied.

I chose to focus on Construction and Manufacturing to keep the study's internal validity strong. In digital business, assets can be changed easily, but this flexibility makes project management very different from industries where changes are permanent. Including IT would have made the analysis too scattered, since software development's trial-and-error approach does not have the same high stakes as heavy industry. By narrowing the focus, I could better study the mental challenges of early planning in fields where mistakes cannot be easily fixed later.

As a result, I did not throw out the IT interview data. Instead, I used it as a secondary comparison. This lets me compare fast, flexible decision-making in IT with the more fixed strategies needed in physical industries. This change kept the analysis focused and helped us see how project managers handle the early stages when they cannot rely on the flexibility of digital projects.

6 Data Analysis

6.1 Risk and Decision-Making in the Construction Sector: Structured Analysis

6.1.1 Segment 1: Early-Stage Risk Focus (The External Dominance)

The data demonstrate that Construction Project Managers predominantly externalise risk. Seventy-five percent (n=9) of participants identified their primary risks as "External factors beyond direct control," including weather, global commodity prices, and regulatory bodies. In contrast, only 25% (n=3) emphasised internal execution risks.

Analysis of these external threats reveals a pronounced geopolitical divide between the European (German and Norwegian) and Indian cohorts.

- Regulatory Paralysis: All German participants (100%, n=5) identified "Regulatory/Compliance" or "Legal Hurdles" as one of their top three risks. By comparison, only 20% (n=1) of Indian participants considered regulation a primary threat.
- Environmental Volatility: In contrast, 80% (n=4) of Indian participants cited "Physical/Environmental" factors, such as monsoons, traffic, and soil density, as a primary threat, whereas only 16% (n=1) of the European group did so.

Statistical Insight: The "Construction Mindset" is geographically determined. In Europe, the project is besieged by law; in India, it is besieged by nature.

6.1.2 Segment 2: Identifying Risks in the First 20% (The Digital-Physical Split)

The analysis of risk identification methodologies indicates an even division within the industry, which is strongly associated with the construction type (Greenfield versus Brownfield or Renovation).

- Physical or intrusive methods are preferred by fifty percent (n=6) of the cohort, who reject exclusively digital planning. These Project Managers require on-site techniques, including physical drilling (probe-openings), early morning site visits, and direct material inspections at entry points. This approach is prevalent in the Renovation and Indian Infrastructure sub-sectors.
- Formal or digital methods are utilized by the remaining fifty percent (n=6), who employ 'Paper-First' strategies. These include digital pre-construction processes such as Building Information Modelling (BIM), Hazard and Operability (HAZOP) workshops, and standardized risk registers. This group consists exclusively of Project Managers operating in highly regulated sectors, including German heavy industry, offshore wind, and metro systems.

Statistical analysis reveals that trust in digital tools is absent among Project Managers overseeing Brownfield or Renovation projects. These professionals consistently emphasize the necessity of physical verification, stating that 'old blueprints lie.

6.1.3 Segment 3: Intuition vs. Tools (The Experience Curve)

The choice between relying on "gut feeling" and "formal tools" is determined by factors such as seniority and liability, rather than individual preference.

- Formal Tool Dependency: Forty-two percent (n=5) of the cohort primarily rely on formal tools. This group is composed exclusively of Junior Project Managers (with less than five years of experience) and Public Sector Project Managers. The data indicate that formal tools function as an "audit shield" for these individuals. Junior Project Managers depend on these tools due to limited experience, while Public Sector Project Managers utilize them to mitigate liability.
- Intuition/Hybrid Reliance: Fifty-eight percent (n=7) of the cohort, all of whom have more than fifteen years of experience, prioritize intuition or a hybrid approach. These experienced professionals characterize intuition as "pattern recognition," such as identifying when a subcontractor is overwhelmed.

Statistical Insight: There is a 100% correlation between "High Liability/Public Scrutiny" roles (e.g., Metro, Offshore, MedTech-Construction) and strict tool adherence. No Project Manager in a liability-heavy role admitted to relying on gut feeling.

6.1.4 Segment 4: Decision-Making with Incomplete Information (Universal Loss Aversion)

This section presents the most statistically significant finding within the Construction dataset.

- Over-Engineering (Buying Certainty): All participants (100%, n=12) responded to incomplete data by allocating additional budget to ensure schedule security.
- Examples of these actions include ordering 25% more rebar, utilizing heavier cranes, purchasing modular capacity, or excavating deeper piles.

Statistical Insight: Status quo bias and loss aversion were observed to be absolute in this context. None of the Construction Project Managers opted to wait for improved data or to attempt a less expensive option initially. The industry consistently incurs additional costs to ensure physical redundancy.

6.1.5 Segment 5: Time and Management Pressure (The Illusion of Activity)

Under managerial pressure, the cohort demonstrated two distinct behavioural patterns:

- The Illusion of Progress: Fifty percent (n=6) of Project Managers reported initiating physical work prematurely, such as commencing excavation before obtaining permits or erecting scaffolding prior to completing surveys, in order to meet management's expectations for visible progress. This behaviour results in a "hidden risk" profile, where the site appears active while the legal and technical foundations remain incomplete.

- Weaponised Compliance: The remaining 50% (n=6), primarily in high-safety sectors such as bridge construction or heavy industry, responded to managerial pressure by strictly adhering to safety protocols. These Project Managers leveraged safety regulations or referenced recent structural failures to compel management to reduce the pace of work.

Statistical Insight: Project Managers in the residential and commercial sectors are three times more likely to exhibit the "Illusion of Progress" behaviour compared to those in infrastructure or heavy industry, who are more inclined to resist such pressures.

6.1.6 Segment 6: Cross-Industry Blind Spots (The Permanence Factor)

Participants identified irreversibility as the primary concept misunderstood by other industries regarding construction.

- Physical/Logistical Constraints: Seventy-five percent (n=9) of participants cited physical realities Weather, Gravity, Soil, and Logistics as the primary blind spot of outsiders. They emphasised that unlike code, physical errors cannot be "patched".
- The so-called "Undo Button" fallacy was noted by 58% (n=7) of participants, who explicitly contrasted construction with IT and Agile methodologies. They observed that iterative processes are not feasible when tasks involve pouring concrete or cutting steel.

Statistical Insight: The "Anti-Agile" sentiment is held by 92% (n=11) of the cohort. Only one participant (Smart Buildings) expressed a desire for more agile-like flexibility, and even they relied on "Over-ordering" hardware to achieve it.

6.1.7 Segment 7: Practical Outcomes (The Wish List)

The solutions requested correspond to the division identified in Segment 3, which distinguishes between physical and digital domains.

- Enhanced Visibility: Fifty percent (n=6) of the cohort requested tools designed to identify previously undetectable physical risks, such as sensors for concrete curing, subsea digital twins, or real-time supply chain tracking.
- Standardisation and Legal Aid: Forty-two percent (n=5), primarily from the German and public sector group, requested standardised contracts or legal stress tests to facilitate navigation of the regulatory environment.

Statistical Insight: Only 17% (n=2) of construction project managers requested soft skills training, while the majority (83%) expressed a preference for more robust data or enhanced legal protection.

Table 2: Risk Perception and Decision-Making in the Construction Sector (n=12)

Category	Sub-Category	Percentage	Description / Insight
Primary Risk Focus	External Factors	**75%**	Focus on weather, regulations, and global prices.
	Internal Execution	**25%**	Focus on team performance and site management.
Identification Method	Physical (Somatic)	**50%**	"Boots on the Ground" verification (drilling, site visits).
	Digital (Formal)	**50%**	"Paper-First" strategies (BIM, Risk Registers).
Tool Reliance	Intuition/Hybrid	**58%**	primarily experienced managers (15+ years).
	Formal Tools	**42%**	Primarily junior managers and public sector roles.
Response to Uncertainty	Over-Engineering	**100%**	Universal willingness to pay a "safety premium" to secure schedule.
Response to Pressure	Illusion of Progress	**50%**	Starting physical work prematurely to look busy.
	Weaponised Compliance	**50%**	Using safety protocols to force a slowdown.
Attitude toward Agile	Anti-Agile	**92%**	Belief that iterative methods are reckless due to physical irreversibility.

6.2 Risk and Decision-Making in the Manufacturing Sector: Structured Analysis

6.2.1 Segment 1: Early-Stage Risk Focus (The Dependency Crisis)

In contrast to the construction sector, where environmental factors are often predominant, the Manufacturing cohort identifies Supply Chain Dependency as the principal risk. The phenomenon referred to as "Scarcity Obsession" is apparent, with 55% (n=6) of participants, particularly those in Electronics, Automotive, and Automation, citing "Component Scarcity" or "Supply Chain Volatility" as their primary concern. This cohort expresses apprehension that the absence of a low-cost component, such as a £1 sensor, could halt an entire multi-million pound production line.

Furthermore, 18% (n=2) of MedTech and Pharma participants identified "Regulatory Compliance" as an existential threat. For these project managers, failure to meet FDA or EU MDR

requirements results in the product being ineligible for legal market entry. Notably, 73% (n=8) of the cohort externalized their risks, perceiving their projects as vulnerable to global factors such as chip shortages and shipping delays, rather than to internal design issues.

A notable statistical insight is that concerns regarding "Global Logistics," including shipping bottlenecks and port closures, were cited exclusively by the Asian cohort (China and India), reflecting their predominant role as exporters. In contrast, the German group focused primarily on risks related to "Specification/Tolerance."

6.2.2 Segment 2: Identifying Risks in the First 20% (Methodological Polarisation)

Approaches to risk identification in manufacturing demonstrate a clear division based on company size and cultural context. Formal methods are prevalent, with 45% (n=5), primarily from large German firms and the High-Tech Electronics sector, utilizing structured tools such as Failure Mode and Effects Analysis (FMEA) or comprehensive Design for Manufacturing audits. These organizations conceptualize risk as a quantifiable variance.

Conversely, 27% (n=3), all from Indian small and medium-sized enterprises (SMEs) and the pharmaceutical sector, prefer somatic methods. These include identifying risks through direct observation, auditory cues from machinery, and assessment of team morale, often prioritizing sensory judgment over digital sensor data. Additionally, 18% (n=2) in the Smartphone and Consumer Electronics sector employ competitor surveillance, monitoring the actions of market leaders to anticipate material shortages and respond proactively.

A statistical analysis reveals a complete correlation between roles in the Heavy Industry and Automotive sectors and the use of formal FMEA templates. No participant in this sub-sector relies exclusively on intuition for risk identification.

6.2.3 Segment 3: Intuition vs. Tools (The Defensibility Gap)

The decision to rely on formal tools rather than intuition is closely associated with perceived liability risks. A majority, 55% (n=6), primarily from the MedTech, Automotive, and Semiconductor sectors, rely on formal tools, considering them essential for legal defensibility in cases of recall or failure. As one participant stated, without a documented paper trail, a risk "does not exist" in the eyes of the regulator.

In contrast, 36% (n=4), primarily from startups and small to medium-sized enterprises (SMEs), prioritize intuition, contending that formal tools are too slow for the pace of their markets. For these participants, risk registers are frequently outdated immediately after documentation. Notably, one participant from the pharmaceutical sector described their risk response as variable and influenced by stress levels, illustrating the Affect Heuristic.

All participants (100%) in the "Junior/Startup" cohort, defined as those with less than five years of experience in startups, rejected the use of formal risk matrices. Instead, they favoured "Daily Stand-ups" or "Flash Syncs" to enable faster and more agile responses.

6.2.4 Segment 4: Decision-Making with Incomplete Information (Buying Options)

When confronted with incomplete information, the Manufacturing sector demonstrates a distinct form of loss aversion, referred to as "Paying for Optionality." Approximately 64% (n=7) allocated capital to secure flexibility, such as ordering machinery with higher specifications or purchasing chips in anticipation of shortages. Participants justified these expenditures by emphasizing that the cost of such "excess" is negligible compared to the potential consequences of a production line stoppage.

In contrast, 18% (n=2) of respondents from the MedTech and Pharma sectors elected to halt projects rather than proceed amid uncertainty, prioritizing safety and regulatory compliance. Another 18% (n=2), mainly from Startups and SMEs, accepted greater risk by engaging unproven suppliers or materials to sustain market momentum.

Notably, the "Defensive Hoarding" strategy, which involves purchasing inventory without confirmed demand, was exclusive to the High-Volume Electronics sector and was reported by all participants in this group.

6.2.5 Segment 5: Time and Management Pressure (The Yield Trap)

Quality assurance frequently deteriorates under management pressure, especially concerning the extended phase of product life. Sixty-four percent (n=7) of project managers reported prioritizing the "Happy Path" or volume targets when under pressure, sacrificing yield (good units) for volume (total units) and accepting increased scrap rates to meet deadlines.

Twenty-seven percent (n=3) indicated that such pressure results in premature design freezes, which may embed defects prior to tooling and cause irreversible flaws. In contrast, only 18% (n=2) of respondents from the MedTech and pharmaceutical sectors actively resisted this pressure by leveraging regulatory requirements to delay or halt progress.

Statistical Insight: A distinct division exists in the concept of "Technical Debt." Project managers in software and electronics intentionally accumulate technical debt to be addressed later through updates or recalls. In contrast, project managers in heavy industry caution that "Physical Debt," such as a flawed engine block, is permanent and cannot be rectified.

6.2.6 Segment 6: Cross-Industry Habits and Blind Spots (The Physics of Failure)

The manufacturing cohort consistently expresses scepticism regarding the application of "Agile" methodologies to physical hardware. Eighty-two percent (n=9) identified "The Physics of Failure" as a primary limitation overlooked by other industries, emphasizing that once metal is cut or plastic is moulded, it cannot be "patched" in the manner of software.

Additionally, 36% (n=4) of respondents in the Tech Hardware sector highlighted the rapid pace of obsolescence, referred to as "Inventory Rot." They noted that electronic components may become obsolete within six months, which contrasts sharply with the extended lifespan of construction materials.

All (100%) participants from the Automotive and Heavy Industry sectors employed the metaphor of "patching" to illustrate why IT-based approaches are ineffective in their field, thereby reinforcing the concept of Physical Permanence.

6.2.7 Segment 7: Practical Outcomes (The Wish List)

The manufacturing cohort primarily identified a need for enhanced visibility and advanced simulation capabilities. Over half of respondents (55%, n=6) requested tools that increase transparency regarding Tier-2 and Tier-3 suppliers, indicating limited trust in direct suppliers and a need for greater insight into the status of raw materials earlier in the supply chain.

Additionally, 36% (n=4) of respondents sought improved simulation tools, such as digital twins, to enable virtual testing of failure modes and thereby reduce the substantial costs linked to physical prototyping. Distinctively, 18% (n=2) requested training in "Emotional Intelligence" or "Conflict Resolution," identifying the high-stress and blame-oriented environment on the factory floor as a major obstacle to effective decision-making.

Table 3: Risk and Decision-Making in the Manufacturing Sector (n=12)

Category	Sub-Category	Percentage	Description / Insight
Primary Risk Focus	Supply Chain Dependency	**55%**	"Scarcity Obsession" focusing on component shortages and volatility.
	Regulatory Compliance	**18%**	Existential threat focus (FDA/EU MDR); primarily MedTech/Pharma.
Identification Method	Formal (FMEA/Audit)	**45%**	Rigorous "Design for Manufacturing" audits; dominant in German firms.
	Somatic (Sensory)	**27%**	"Walking the floor" and listening to machines; dominant in Indian SMEs.
	Competitor Surveillance	**18%**	Monitoring market leaders to anticipate shortages.
Tool Reliance	Formal Tools	**55%**	Used for legal "Defensibility" in case of recalls.
	Intuition	**36%**	Rejected formal matrices as too slow; dominant in Startups.
Response to Uncertainty	Paying for Optionality	**64%**	Spending capital on inventory/flexibility to avoid line stoppages.
	Halt Project	**18%**	Prioritizing safety over progress (MedTech).
Response to	The Yield Trap	**64%**	Trading yield (quality) for volume to

Pressure			meet shipping targets.
	Premature Freezing	**27%**	Locking in defects before tooling to save time.
Attitude toward Agile	Physics of Failure	**82%**	Rejection of "patching" mentality; physical errors cannot be undone.

6.3 Risk and Decision-Making in the IT (Information Technology) Sector: Structured Analysis

6.3.1 Segment 1: Early-Stage Risk Focus (The Dependency Trap)

The risk landscape in the IT sector differs significantly from that of Construction and Manufacturing. Project Managers (PMs) in IT are primarily concerned with interdependency challenges and maintaining relevance in a rapidly evolving digital environment. 67% (n=4) of the cohort identified "Third-Party Failure" or "Integration Issues," including unstable APIs, changing cloud provider terms, or legacy system conflicts, as their principal risks. In contrast to physical assets, software modules are inherently dynamic and may fail unexpectedly due to external changes beyond the team's control. For half of the group (n=3), particularly those in Fintech and Banking, "Regulatory Compliance" (GDPR, BaFin, DORA) constitutes an existential threat, rendering legal oversight as critical as technical execution. Additionally, 33% (n=2), mainly from Startups and SaaS, highlighted "Market Timing" or "Scope Creep" as primary concerns, where the predominant risk lies in developing an unsuitable product or missing market opportunities. Notably, none of the IT PMs identified physical or raw material risks; instead, the predominant concerns are logical, legal, or temporal rather than tangible.

6.3.2 Segment 2: Identifying Risks in the First 20% (Simulation vs. Inspection)

In contrast to Construction and Manufacturing project managers, Information Technology project managers cannot perform physical site walks to identify risks. Instead, 50% (n=3) utilize pre-mortems or threat modelling, which are cognitive exercises that require teams to envision project failure and work backward to identify potential unknowns. An additional 33% (n=2) employ technical spikes or proof of concept builds, developing disposable code to rapidly test high-risk assumptions. Furthermore, 50% (n=3), primarily in large corporate environments, depend on gap analysis or alignment workshops to reveal friction between legacy and new systems. These approaches are inherently collaborative and code-based, as visual inspection is not applicable when the project site is an abstract codebase rather than a physical location.

6.3.3 Segment 3: Intuition vs. Tools (The Maturity Split)

Reliance on tools within the IT sector is strongly associated with organisational maturity and the level of regulatory oversight. 33% (n=2) of the cohort, comprising Banking and Junior SaaS Product Managers (PMs), rely almost exclusively on formal tools for audit and validation, as

professional experience alone is insufficient in environments with stringent compliance requirements. In contrast, 17% (n=1), represented by the Startup PM, primarily depends on intuition, contending that rapid market fluctuations often render data obsolete before analysis is complete. The majority (50%, n=3) employ a hybrid strategy, utilising tools for technical tracking while relying on intuition for people management, particularly to identify issues such as team burnout or stakeholder resistance. In contrast to physical industries, intuition in IT primarily serves as an indicator of social rather than product health.

6.3.4 Segment 4: Decision-Making with Incomplete Information (Buying Modularity)

When confronted with uncertainty, IT project managers (PMs) predominantly invest in optionality by over-engineering modular architectures. 83% (n=5) select more expensive or complex solutions, such as Multi-Cloud, Modular Design, or Wrapper APIs, to maintain the ability to adapt as requirements change. This approach is based on the belief that, although code can be modified, the underlying architecture is less adaptable; therefore, flexibility is prioritized over simplicity. In contrast, 17% (n=1), specifically the SaaS PM, intentionally selects the fastest and least expensive approach ("Sandboxing") to identify flaws early, as the primary cost of failure is time rather than materials. All Corporate and Fintech PMs (100%) prefer over-engineering to ensure security and redundancy, considering regulatory penalties to be more significant than development overruns.

6.3.5 Segment 5: Time and Management Pressure (The Technical Debt Trade-off)

Time pressure in information technology (IT) environments frequently results in the accumulation of technical debt, as short-term decisions introduce concealed, long-term risks. 83% of participants (n=5) reported prioritizing visible features, such as user interface elements, over backend integrity or documentation to satisfy management and achieve rapid results. Half of the group (n=3) specifically cautioned that security audits and integration tests are often omitted under deadline constraints, thereby introducing latent risks that only become apparent after deployment. The E-commerce project manager described the implementation of code freezes during critical periods, such as Black Friday, acknowledging that increased agility can paradoxically lead to greater fragility when the consequences of errors are most severe. In contrast to their counterparts in construction, who are concerned with physical collapse, IT project managers primarily fear a maintenance collapse, in which a disorganized codebase becomes impossible to update safely.

6.3.6 Segment 6: Cross-Industry Habits and Blind Spots (The Update Fallacy)

The primary cross-industry misunderstanding identified by IT project managers is the "Patching Mentality." 67% (n=4) report that external stakeholders significantly underestimate the fragility of software dependencies, often assuming that software functions as a standalone asset. In reality, updates from third-party vendors can disrupt entire systems. Additionally, one-third (33%, n=2) highlight the "90% Done" misconception, where a system appears nearly complete due to a finished user interface, while the most complex backend logic remains unaddressed. The banking project manager further identifies a "Cyber-Physical" misunderstanding: non-IT

stakeholders often conceptualize security as a locked door, such as a firewall, whereas IT professionals recognize that the primary risk involves lateral movement within systems after initial access. These findings emphasize the distinct and often invisible risks inherent in virtual systems.

6.3.7 Segment 7: Practical Outcomes (The Wish List)

Improvement requests from IT project managers primarily emphasize automation and system design rather than legal frameworks. 67% (n=4) identified a need for enhanced automated testing, continuous compliance, or real-time analytics to detect issues early and minimize human error in risk management. An additional 33% (n=2) requested the implementation of risk budgets, defined as time and resources allocated for unforeseen events, or advocated for more modular architectures to limit the consequences of inevitable failures. Notably, none of the IT respondents requested improved contracts or legal training, which are common priorities in the construction sector. Instead, they prioritized improved protocols and system architecture as the most effective means of risk reduction.

Table 4: Risk and Decision-Making in the IT Sector (n=6)

Category	Sub-Category	Percentage	Description / Insight
Primary Risk Focus	Third-Party Failure	**67%**	Focus on integration issues (unstable APIs, cloud provider changes).
	Regulatory Compliance	**50%**	Existential threat focus (GDPR, DORA); primarily in Fintech/Banking.
	Market Timing	**33%**	Fear of "Scope Creep" or arriving too late; primarily Startups/SaaS.
Identification Method	Cognitive (Pre-Mortem)	**50%**	"Threat Modelling" and imagining failure backwards.
	Technical Spikes	**33%**	Creating disposable "Proof of Concept" code to test assumptions.
	Gap Analysis	**50%**	Workshops to align legacy systems with new requirements.
Tool Reliance	Hybrid (Social Intuition)	**50%**	Intuition is used to gauge **team burnout**, tools used for code tracking.
	Formal Tools	**33%**	Exclusive reliance for audit/validation (Banking).
Response to Uncertainty	Buying Modularity	**83%**	"Over-engineering" architecture (e.g., Multi-Cloud) to preserve flexibility.
	Sandboxing	**17%**	Choosing the cheapest/fastest path

			to fail fast (SaaS).
Response to Pressure	Technical Debt	**83%**	Prioritizing visible features (UI) over backend integrity to appease stakeholders.
	Skipping Tests	**50%**	Bypassing security audits or integration tests to meet deadlines.
Cross-Industry Blind Spot	The Patching Mentality	**67%**	Outsiders underestimate the fragility of software dependencies.

6.4 Cross-Industry Horizontal Analysis: Project Management DNA Across Construction, Manufacturing, and IT

6.4.1 Segment 1: Early-Stage Risk Focus (The Taxonomy of Fear)

An analysis of risk focus at the inception of projects reveals that all three sectors exhibit a pronounced "External Locus of Control," characterized by the belief that the most significant threats originate beyond the team's direct influence. The specific nature of these external threats, however, differs fundamentally across sectors.

- Construction (The Hostile Environment): Seventy-five percent (n=9) of construction project managers identify the physical environment, including weather, soil conditions, and regulatory challenges, as their primary concern. Projects in this sector are perceived as objects exposed to a hostile and unpredictable environment.
- Manufacturing (The Hostile Market): Fifty-five percent (n=6) of manufacturing project managers express concern about disruptions to inputs, such as supply shortages or logistics failures. In this context, projects are regarded as systems dependent on the consistent availability of components and raw materials.
- IT (The Hostile Network): Sixty-seven percent (n=4) of IT project managers identify networked dependencies, such as unstable application programming interfaces (APIs) and integration challenges, as their primary concern. IT projects are considered highly susceptible to risks arising from disconnection or failure within interconnected systems.

Insight: Regulatory and compliance risk is the only universally cited concern, with 100% of German construction, 18% of manufacturing (MedTech/Pharma), and 50% of IT (Fintech/Banking) project managers identifying it as a top threat. This finding suggests that, across all sectors, legal requirements exert a more significant influence than either physical or technological factors.

6.4.2 Segment 2: Identifying Risks in the First 20% (Somatic vs. Cerebral)

This section highlights significant differences in the epistemology of risk identification. Physical industries such as Construction and Manufacturing prefer somatic verification.

- The "Touch" Standard: 48% (n=14) of Construction and Manufacturing participants rely on sensory input, such as listening to machines, feeling vibrations, or observing materials in situ. These participants tend to mistrust abstract data and instead prioritize direct physical evidence.
- The "Mind" Standard: 100% (n=6) of IT project managers employ cognitive approaches such as simulation, pre-mortems, and technical spikes to identify and assess risks in intangible systems where physical inspection is not applicable.

Despite methodological differences, both the construction project manager inspecting for a "Made in India" stamp and the IT project manager conducting a technical spike engage in a form of "proof-of-reality." Both groups exhibit scepticism toward documentation and prioritize empirical validation over theoretical representations.

6.4.3 Segment 3: Intuition vs. Tools (The Liability Shield)

Among all 30 participants, reliance on formal tools is primarily influenced by concerns regarding personal liability rather than by the specific industry sector.

- The "Audit Shield" cluster comprises high-liability roles, including Metro Tunnels, MedTech, and Banking, all of which rely entirely on formal tools and documented processes. Participants in these roles assert that intuition does not provide legal protection.
- The "Survival" cluster includes participants in high-speed, low-documentation roles, such as Startups and SMEs, who avoid formal tools because they consider them too slow for their rapidly changing environments.

The concept of "intuition" varies by sector. In Construction and Manufacturing, intuition is associated with physical pattern recognition in materials and processes. In contrast, in Information Technology, intuition is primarily social, involving the detection of team burnout or stakeholder resistance. Notably, 50% of IT project managers use intuition for people management, whereas Construction project managers focus exclusively on material intuition.

6.4.4 Segment 4: Decision-Making with Incomplete Information (The Universal Safety Premium)

Across all sectors, the overwhelming majority, 93% (n=27), respond to uncertainty by allocating capital to increase certainty.

- Construction: Increases material usage, such as additional rebar or deeper foundations.
- Manufacturing: Accumulates inventory, for example by stockpiling components or specifying machinery beyond requirements.

- Information Technology: Invests in modularity, such as implementing multi-cloud set-ups or utilizing wrapper APIs.

This behaviour reveals a universal "Psychological Insurance Premium." Regardless of whether additional concrete is used or redundant architecture is implemented, the underlying motivation is loss aversion. Project managers prefer to justify overspending as "prudence" rather than risk project delays or perceptions of incompetence.

6.4.5 Segment 5: Time and Management Pressure (The Illusion of Progress)

When operating under time constraints, a recurring behavioural pattern known as the "Potemkin Village Effect" emerges. Project managers often prioritize the appearance of progress to satisfy management, which can undermine actual project health.

- In construction, 50% of projects commence visible activities such as digging or erecting scaffolding before obtaining necessary permits, thereby prioritizing immediate action over regulatory compliance.
- Within manufacturing, 64% of operations emphasize shipping volume, occasionally neglecting product yield or long-term quality considerations.
- In information technology, 83% of teams prioritize frontend or user interface development, deferring backend functionality and security integrity to later stages.

Information technology is particularly susceptible to the Potemkin Village Effect (83% frequency), likely because technical debt and latent risks are less visible compared to the tangible defects present in physical industries.

6.4.6 Segment 6: Cross-Industry Blind Spots (The Physics of Reversibility)

The primary cross-industry distinction concerns reversibility, specifically the cost and feasibility of correcting errors.

- The "One-Shot" mindset is prevalent among Construction and Manufacturing project managers, with 87% (n=20) identifying irreversibility, such as concrete setting or metal cutting, as the primary risk. In these sectors, methodologies like Agile are perceived as reckless rather than adaptive.
- The "Patching" mindset characterizes IT project managers, who conceptualize risk in temporal terms and are comfortable with early launches followed by subsequent corrections, given that software errors are theoretically reversible.

This distinction accounts for the mutual scepticism between sectors. Manufacturing project managers regard IT’s iterative approach as hazardous, whereas IT project managers perceive the emphasis on physical perfectionism as slow and inflexible. Essentially, one group prioritizes prevention of errors, while the other emphasizes rapid correction.

6.4.7 Segment 7: Practical Outcomes (The Wish List)

Despite sectoral differences, all industries share a common objective: achieving greater visibility through tools that reveal previously unrecognized layers of risk.

- The construction sector seeks enhanced underground insight, exemplified by the use of soil sensors.
- The manufacturing sector prioritizes upstream visibility, such as achieving transparency with Tier-2 suppliers.
- The information technology sector emphasizes internal clarity, for example through the implementation of automated code analytics.

Comparative Analysis

- Legal/Standardisation Requests: Construction (42%) > Manufacturing (18%) > IT (0%). This suggests Construction is particularly dispute-prone, while IT focuses on design. Soft skills training is most prevalent in manufacturing (18%), followed closely by construction (17%), and is not observed in information technology (0%). The heightened need for emotional resilience in manufacturing and construction is driven by the physical danger and stress associated with shop floor and site environments, a concern that is less significant in the information technology sector. cern less pronounced in the IT sector.

Table 5: Cross-Industry Horizontal Analysis (n=30)

Category	Sub-Category	Percentage	Description / Insight
Primary Risk Focus	**The Hostile Environment** (Construction)	**75%**	Project is an object subjected to weather and soil.
	The Hostile Market (Manufacturing)	**55%**	Project is a machine vulnerable to supply scarcity.
	The Hostile Network (IT)	**67%**	Project is a node vulnerable to API/Integration failure.
Identification Method	**Somatic Verification** (Physical)	**48%**	Relying on senses (touch/sound); dominant in Construction & Mfg.
	Cerebral Simulation (Virtual)	**100%**	Relying on Pre-Mortems/Spikes; exclusive to IT.
Tool Reliance	**The Audit Shield**	**100%**	In high-liability roles (Metro, MedTech, Banking), tools are used strictly for legal defence.
	Survival Mode	**100%**	Startups/SMEs reject tools as "too slow".
Response to Uncertainty	**The Safety Premium**	**93%**	Universal willingness to overspend to buy certainty (Mass,

			Inventory, or Modularity).
Response to Pressure	**The Potemkin Village**	**83%**	Prioritizing visible progress (UI/Digging) over invisible quality to satisfy stakeholders.
Cross-Industry Blind Spot	**Irreversibility**	**87%**	Construction/Mfg. see "Agile" as reckless because physical errors cannot be "patched".
	Reversibility	**67%**	IT sees "Perfectionism" as slow because code can always be fixed later.

6.5 Universal PM DNA: Foundational Similarities Across Project Managers

6.5.1 The Strategic Redundancy Reflex (Radical Loss Aversion)

- Statistical Frequency: 90% (n=27/30).
- Behavioural Synthesis: The most statistically significant convergence observed across the cohort pertains to responses to "Incomplete Information." When confronted with unknown variables, project managers typically do not pause to seek clarity. Instead, they allocate capital proactively as a protective measure. Participants consistently favor "Over-Specification" or "Early Procurement" of non-essential flexibility rather than risk project idleness. The data indicate that project managers regard "Budget" as a renewable resource, while "Schedule" and "Political Reputation" are considered finite. This pattern suggests that the potential cost of waste, such as ordering 25% extra material or purchasing 15% more cloud capacity, is viewed as a more professional and "defensive" approach than risking a project stoppage.
- Analytical Driver: This behaviour exemplifies loss aversion. The negative impact of a project stoppage outweighs the professional discomfort associated with a budget overrun, resulting in a universal "Safety Premium" regardless of whether the project manager is overseeing construction materials or software development.

6.5.2 Methodological Scepticism (The Social-Technical Verification)

- Statistical Frequency: 93.3% (n=28/30).
- Behavioural Synthesis: Although sophisticated digital dashboards such as BIM, FMEA, and Jira are available, a widespread trust deficit persists regarding formal data. During the initial 20% of the project lifecycle, nearly all participants circumvent formal reporting structures to engage in what may be described as analog verification. This process includes personal site inspections, brief synchronization meetings, or direct consultations with experts. The prevailing view is that the digital plan constitutes a polite fiction, concealing the team's underlying technical or environmental concerns.
- Analytical Driver: Project managers consistently report that human intuition is uniquely

capable of identifying underlying issues that software may overlook. Consequently, project success is associated with a manager's ability to triangulate digital data with social cues from technical leads and experienced shop-floor personnel.

6.5.3 The Optics Bias (Impression Management under Pressure)

- Statistical Frequency: 83.3% (n=25/30).
- Behavioural Synthesis: When management or stakeholder pressure increases, the data indicate a consistent breakdown of the "Iron Triangle" (Time, Cost, Quality) that follows a specific pattern: "Invisible Quality" is often sacrificed to maintain "Visible Progress." Under such conditions, project managers prioritize milestones that fulfil the visual or administrative expectations of stakeholders, such as initiating groundwork without permits, delivering lower-yield hardware, or developing high-fidelity user interfaces on unstable backends. This "Potemkin Village" effect serves to manage stakeholder anxiety and secure continued funding, even though it may result in significant "Technical or Structural Debt" that requires remediation in later project stages.
- Analytical Driver: This phenomenon exemplifies Defensive Decision Making through Impression Management. Project managers assess that maintaining the appearance of project progress secures their continued involvement, thereby enabling them to address underlying quality issues during subsequent project phases.

6.5.4 The Externalization of Fatal Risks (Locus of Control)

- Statistical Frequency: 86.7% (n=26/30).
- Behavioural Synthesis: Project managers demonstrate a notable similarity in categorizing existential threats. Although they allocate most of their time to managing internal teams, project managers consistently identify primary threats to project completion as externalities. These externalities encompass regulatory changes, environmental volatility such as weather or soil conditions, and third-party dependencies including APIs and supply chains. The prevailing psychological stance is characterized by a siege mentality, in which project managers perceive themselves as competent defenders confronting a hostile and unpredictable environment.
- Analytical Driver: This shared perspective suggests that project failure is seldom attributed to inadequate internal planning, but rather to the vulnerability of external dependencies. This dynamic accounts for the dominance of the "Insurance Reflex" (Trait 1), as project managers allocate budgetary resources to safeguard projects against factors perceived as beyond their control.

6.5.5 Defensive Tooling (Liability Shielding)

- Statistical Frequency: 63.3% (n=19/30).
- Behavioural Synthesis: The use of formal risk management tools correlates more strongly with perceived legal or professional liability than with the technical complexity of a project. In high-consequence sectors such as banking, public infrastructure, and medical manufacturing, tools like the "Risk Register" or "VOB" function primarily as

bureaucratic safeguards. Participants in these industries report that if a risk is not documented officially, the project manager assumes personal liability. In contrast, in high-velocity sectors with lower immediate liability, such as startups and warehousing, these tools are often perceived as impediments and are frequently disregarded.

- Analytical Driver: These findings indicate that formal methodologies are frequently employed as mechanisms for liability mitigation rather than for genuine project optimization. The professional approach of project managers shifts from intuitive decision-making to strict adherence to protocol as the potential cost of failure rises due to increased litigation risk.

6.5.6 Strategic Synthesis for Research Implications

- The research indicates that the "Human Element" in project management is not limited to any specific industry. To enhance success rates, organizational training should shift focus from industry-specific technical tools to the development of Uncertainty Competency.
- Given that 90% of managers are likely to waste budget to avoid anxiety and 83% may compromise long-term quality for short-term appearances, the primary "Success Factor" is not the use of improved project management tools, but rather the psychological resilience of the project manager and the establishment of an organizational culture that supports the "Strategic Pause" instead of the "Expensive Guess."
- This analysis demonstrates that although the specific materials and contexts of various industries may change, the underlying behavioural architecture of decision-makers remains consistent.

Table 6: Universal PM DNA - Foundational Similarities (n=30)

Trait / Behaviour	Sub-Category	Frequency	Description / Insight
1. The Strategic Redundancy Reflex	Radical Loss Aversion	**90.0%**	Managers prioritize "Over-Specification" or early procurement to "buy certainty." They view budget as a renewable resource, but time and reputation as finite.
2. Methodological Scepticism	Social-Technical Verification	**93.3%**	A universal "Trust Deficit" toward formal digital reports. Managers prefer "Analog Verification" (site walks, direct huddles) to find the "hidden truth" that software misses.
3. The Optics Bias	Impression Management	**83.3%**	Under pressure, "Invisible Quality" is sacrificed for "Visible Progress" (e.g., the Potemkin Village effect) to manage stakeholder anxiety and secure funding.
4. Externalization of Fatal Risks	Locus of Control	**86.7%**	A "Siege Mentality" where managers view the primary threats as "Externalities" (regulators, weather, supply chains) rather than internal planning failures.

5. Defensive Tooling	Liability Shielding	**63.3%**	In high-liability roles, formal tools (Risk Registers) are used primarily as a "Bureaucratic Shield" for personal protection against litigation rather than for project optimization.

6.6 Methodological Refinement: Excluding the IT Sector for Analytical Clarity

To preserve internal validity, the study deliberately excluded the IT/Software sector from the final comparative analysis. While initial horizontal analysis identified common psychological reflexes among project managers, subsequent data indicated that IT comprises a fragmented set of distinct subfields rather than a unified industry. Including IT would likely introduce statistical noise, thereby reducing the clarity and relevance of the findings for physical project management.

6.6.1 The Paradox of Internal Variance in IT

- The analysis indicated that variance within the IT cohort significantly exceeded the variance observed between the combined Construction and Manufacturing sectors. Whereas physical industries generally share consensus regarding risks such as supply chains and environmental factors, IT participants demonstrated a wide range of divergent philosophies.
- The Agile Conflict: For project managers in banking infrastructure, the practice of "failing fast" may lead to regulatory failures, while for project managers in Software as a Service (SaaS) startups, it serves as a critical learning mechanism.
- The Stability Split: Certain IT sub-sectors implement strict code freezes, mirroring conservative practices found in manufacturing, whereas others adopt continuous iteration, comparable to constructing an aircraft during flight.

Aggregating these diverse practices under a single "IT" designation is academically misleading, as the sector does not possess a unified approach to project management. Each subsector operates according to distinct rhythms and philosophies, shaped by unique rules and constraints.

6.6.2 The Physics of Irreversibility: The Undo Button Divide

- A fundamental distinction between virtual and physical projects is the concept of reversibility, defined as a project's capacity to reverse or undo decisions.
- Material Permanence: In construction and manufacturing, 87% of participants identified irreversibility as a primary risk. Physical actions, such as pouring concrete or cutting steel, are final and impose considerable psychological pressure because of their permanence.
- Virtual Refactoring: In information technology, 83% of project managers actively manage technical debt, frequently releasing imperfect code with the intention of subsequent refactoring or patching. The capacity to delete or modify digital work fundamentally changes the associated risk landscape.

This disparity in reversibility leads to a misleading equivalence between physical and virtual project management, as the psychological factors influencing project managers in each domain are fundamentally different.

6.6.3 Statistical Divergence in Risk Identification

Risk identification methodologies within the information technology sector remain highly fragmented. In contrast, the Construction and Manufacturing sectors demonstrate 75–90% consensus regarding major stressors. IT sector responses are distributed among audit frameworks, intuitive synchronization meetings, and load testing procedures. The lack of a dominant methodology in IT hinders the development of a standardized risk model that could substantially inform this research.

6.6.4 Conclusion: Strategic Focus on Physical Determinism

- Excluding the IT sector allows this study to focus on the "Physics of Irreversibility," facilitating a rigorous comparative analysis of the construction and manufacturing industries. These related sectors are characterized by the following:
- Physical Permanence: The significant cost and finality associated with correcting errors.
- Supply Chain Dependence: The volatility inherent in transporting physical goods.
- Environmental Friction: The challenges resulting from gravity, weather conditions, and material fatigue.

This refined scope ensures that the thesis focuses on a stable, high-stakes environment in which project management is determined by unyielding physical laws, as opposed to the flexible and iterative processes characteristic of virtual code.

Table 7: Methodological Refinement - Exclusion of the IT Sector

Rationale Category	Key Difference / Insight	Description
Internal Variance	The "Agile Conflict"	IT is not a unified industry but a fragmented collection of subfields. Practices range from "failing fast" (Startups) to strict code freezes (Banking), creating too much statistical noise for comparison.
Reversibility	The "Undo Button" Divide	**83%** of IT managers consciously manage "Technical Debt" (shipping imperfect code to fix later). In contrast, **87%** of Construction/Mfg. managers cite "Irreversibility" (e.g., concrete setting) as their defining risk.
Risk Identification	Methodological Fragmentation	unlike the 75-90% consensus found in physical industries, IT risk methods are scattered across audit frameworks, intuitive syncs, and load testing, preventing a standardized model.

Ontological Divergence	Virtual vs. Physical	The "fail-fast" nature of software does not carry the catastrophic cost-of-error found in heavy industry. Comparing "Virtual Refactoring" to "Physical Permanence" creates a false equivalency,.

6.7 How Experience Shapes Risk Perception: Insights from 24 Construction and Manufacturing Project Managers (Macro & Micro)

6.7.1 The Experience Filter – How Seniority Shapes the Eye

In project management, particularly within the construction and manufacturing sectors, the identification and prioritization of risks are significantly shaped by the manager's years of experience. Analysis of data from 24 participants reveals distinct patterns in risk perception corresponding to different stages of a project manager's career.

1.1.1.5 The Junior Perspective (Micro): The Small Detail Trap (1–5 Years)

Project managers in the early stages of their careers (1–5 years) demonstrate a pronounced focus on technical risks. Due to limited experience with project failures and recoveries, junior managers often perceive minor technical errors as significant threats. The study found that 100% (n=8) of junior project managers identified technical failures, such as faulty sensors or unusual soil conditions, as their primary concern. Their approach is predominantly procedural, relying on manuals and checklists to address the complexities of their roles. This hyper-vigilance ensures rigorous monitoring of physical details on-site, but it frequently results in reduced awareness of broader risks, including regulatory changes or contract ambiguities.

1.1.1.6 The Senior Perspective (Macro): The Big Picture Blindness (20+ Years)

In contrast, project managers with more than 20 years of experience tend to focus less on technical details and more on structural and legal risks. All senior participants in the study (100%, n=5) excluded "Technical Errors" from their primary risk concerns, instead emphasizing challenges such as "Government Regulations" and "Contract Law." These experienced professionals approach project management from a strategic perspective and demonstrate proficiency in navigating political and financial environments. Nevertheless, prolonged exposure to recurring technical issues may result in a diminished awareness of minor but potentially significant errors at the operational level, as they have become accustomed to such problems being resolved without major consequences.

1.1.1.7 Getting Used to Errors: The Veteran's Trap

A key finding from the research is the phenomenon of "Normalization of Small Risks." Senior project managers, having witnessed minor safety breaches over extended periods without adverse outcomes, may no longer perceive these incidents as significant threats and instead accept them as routine. In contrast, junior project managers, who are less accustomed to disregarding such breaches, are more likely to identify and report immediate, tangible dangers. This divergence underscores the complementary strengths of both groups: senior managers are proficient in mitigating external threats such as legal actions and budgetary pressures,

whereas junior managers are more effective at detecting internal hazards, including technical oversights and site errors.

1.1.1.8 Bridging the Experience Gap: Towards Balanced Risk Management

Effective project management within physical industries requires balancing overarching strategic objectives with attention to operational details. The study recommends implementing "Reverse Audits," a process in which junior project managers examine operational specifics while senior managers provide guidance on strategic issues. This collaborative method promotes comprehensive risk management, addressing both immediate technical vulnerabilities and broader legal or political risks. Harnessing the complementary strengths of experienced leaders and less-experienced team members helps bridge the experience gap and protects projects from the diverse risks present in high-stakes environments.

Table 8: How Experience Shapes Risk Perception (n=24)

Category	Junior Perspective (Micro)	Senior Perspective (Macro)	Core Insight / Contrast
Experience Level	**1–5 Years**	**20+ Years**	The "Experience Gap" creates two distinct realities within the same project.
Primary Risk Focus	**Technical Failure**	**Structural & Legal Risks**	Juniors fear the machine breaking; Seniors fear the contract breaking,.
Specific Examples	Faulty sensors, soil density, unusual vibrations.	Government regulations, contract ambiguities, financial variance.	
Cognitive Approach	**Hyper-Vigilance**	**Normalization**	Juniors treat every small error as a potential disaster; Seniors often dismiss small errors as "routine noise".
Methodology	**Procedural**	**Strategic**	Reliance on manuals and checklists to navigate complexity.
Blind Spot	**Strategic Blindness**	**The Veteran's Trap**	Juniors miss broader regulatory changes.

6.8 Risk Management Archetypes: A Comparative Analysis of German and Indian Project Managers

This section examines two distinct archetypes in project management: the Process Guardian, exemplified by German project managers, and the Resource Navigator, commonly observed among Indian project managers. Through cross-cultural analysis, differences in digital tool usage, environmental adaptability, and training requirements are identified, providing guidance for developing more balanced and resilient risk management practices.

6.8.1 Digital Tool Reliance: Shield versus Suggestion

Germany: The "Shield" Approach

- German project managers exhibit a strong reliance on digital compliance and simulation tools, with 91% (n=10 of 11) considering tools such as FMEA and BIM dashboards indispensable. These digital systems serve as both legal and professional safeguards, forming the foundation of their risk mitigation strategies. The stability of German infrastructure further encourages project managers to depend almost exclusively on digital plans, with risks often considered absent unless identified by the software.

India: The "Suggestion" Approach

- Although Indian project managers have access to similar digital tools, 92% (n=12 of 13) utilize them as advisory aids rather than as definitive authorities. The unpredictable nature of the local environment necessitates ongoing human verification. While digital dashboards offer guidance, persistent risks such as monsoons or power outages remain significant factors that extend beyond digital monitoring.

6.8.2 The Jugaad Factor: Managing Environmental Volatility

Indian project managers tend to internalize environmental unpredictability as a primary risk, whereas German project managers typically consider such factors as secondary. The Indian approach, commonly termed "Jugaad," emphasizes resourceful and socio-technical ingenuity to address institutional failures. For instance, in the event of a supply chain collapse, a German project manager may escalate the issue through contractual channels, while an Indian project manager is more likely to utilize personal networks or implement cost-effective workarounds to maintain project momentum. Consequently, Indian project managers are highly adaptive and reactive, demonstrating proficiency in rapid, improvised solutions, whereas German project managers excel at developing comprehensive plans under stable and predictable conditions.

6.8.3 Reverse Training Opportunities: Bridging the Blind Spots

The findings challenge the assumption of universal superiority of Western methods. Each archetype demonstrates distinct strengths and limitations that may be mitigated through targeted cross-training.

For Germany: Crisis Adaptability (Learning from India)

- German project managers may experience operational paralysis when digital protocols fail to address emergent disruptions. Incorporating training modules in heuristic decision-making, modelled on the Indian "Jugaad" approach, could enhance their capacity to manage unforeseen environmental challenges without exclusive reliance on digital infrastructure.

For India: Digital Standardisation (Learning from Germany)

- Indian project managers frequently allocate substantial resources to manual verification and depend extensively on intuition, which may impede scalability and auditability. Training in stringent digital accountability could facilitate a transition from relationship-based to data-driven trust, thereby promoting more consistent and scalable project outcomes.

Table 9: Risk Management Archetypes - Cross-Cultural Analysis (Germany vs. India)

Feature	**German Archetype: "The Process Guardian"**	**Indian Archetype: "The Resource Navigator"**	**Core Insight / Contrast**
Primary Mental Model	**The Shield**	**The Suggestion**	Germans use tools to *protect* against liability; Indians use tools to *guide* through chaos.
Digital Tool Reliance	**91% (n=10/11)**	**8% (n=1/13)**	Germans view the digital plan as the authority; Indians view it as advisory only.
Response to Volatility	**Escalation (Contractual)**	**Jugaad (Improvisational)**	Germans rely on the contract to solve problems; Indians rely on social networks and frugal innovation.
Underlying Fear	**Regulatory Paralysis**	**Environmental Chaos**	Fear of non-compliance (Law) vs. Fear of physical disruption (Nature/Logistics).
Blind Spot	**Rigidity**	**Scalability**	Germans struggle when the "Perfect Plan" fails; Indians struggle to scale their improvisational success.

7 Conclusion

This final chapter provides a strategic framework for current and aspiring project managers. Synthesis of research involving 24 professionals in Construction and Manufacturing reveals that, although the specific materials and contexts may differ, the psychological characteristics of effective managers remain consistent.

7.1 Summary of Research Findings

In conclusion, this study revisits the primary research questions and provides a summary of the evidence collected.

- RQ1: Early-Stage Risk Identification: Managers in both sectors exhibit an external locus of control, perceiving the most significant threats as originating outside the project. In construction, these threats are characterized by environmental factors such as weather and soil conditions, while in manufacturing, market scarcity and supply chain volatility are predominant concerns.
- RQ2: Decision-Making Patterns: When confronted with uncertainty, 93% of project managers opt to pay a safety premium. They prefer to invest in additional materials or higher-specification equipment at the outset rather than risk project delays or stoppages in later stages.
- RQ3: Tools and Techniques: Formal tools like FMEA are used as a "Bureaucratic Shield" in high-liability roles to deflect personal blame. However, most veterans still rely on "Somatic Verification"—physically seeing or hearing the work to find the "hidden truth" that digital dashboard
- RQ4: Improving Success Rates: Project success depends less on advanced software and more on perspective pairing. Integrating the technical insights of junior staff with the strategic and legal expertise of senior personnel enables organizations to address the full spectrum of project risks more effectively.

7.2 A Guide for New Project Managers: Early Risk Identification

As a new project manager in these sectors, allocate the initial 20% of the project timeline to developing a comprehensive risk perspective using the following two approaches:

- Framework (Entropy versus Precision): In construction, the primary role involves managing disorder to establish operational order. In manufacturing, the focus shifts to optimizing precision and preventing minor deviations from affecting large production volumes.
- Habit (Somatic Truth): Digital reports may not always reflect on-site realities. Develop the practice of conducting site visits or directly monitoring equipment. Sensory observations are often the most reliable means of risk detection, as physical materials provide direct evidence.

7.3 The Industry Transition Playbook: Strategies for Reducing Errors

Professionals transitioning between the construction and manufacturing sectors often encounter significant unfamiliarity. The following research-based strategies are intended to facilitate a smoother transition

1.1.1.9 Leverage Core Project Management Competencies (Consistent Elements):

- Loss Aversion: Apply established risk mitigation strategies to ensure schedule reliability across both industries.
- Defensive Documentation: Consistently maintain thorough documentation, as lack of records in either sector may result in personal accountability for associated risks.

1.1.1.10 Close the "Knowledge Gap" (What you must learn):

- Transitioning to Manufacturing: Shift your primary concerns from environmental factors to systemic issues. Instead of weather-related disruptions, focus on risks such as inventory degradation and the compounding effects of minor technical errors.
- Transitioning to Construction: Shift your primary concerns from machinery to legal considerations. The construction industry is significantly more prone to disputes. Develop expertise in contractual law and regulatory compliance, as legal frameworks are foundational to industry operations.

7.4 Closing Statement

This research demonstrates that success is fundamentally a human-centric achievement. Regardless of whether a project involves constructing a bridge in Munich or establishing a production line in Mumbai, the most significant risks often stem from cognitive blind spots rather than material selection. Adopting a hybrid mindset that values both digital data and physical intuition enables contemporary project managers to navigate the initial phases of any project with the confidence necessary to achieve successful completion.

Project management excellence emerges at the intersection of established experience and innovative perspectives, as well as the integration of digital tools with practical, real-world insights.

8 Recommendation

The results of this research indicate that achieving project success in heavy industries requires addressing the human disconnect among different generations, cultures, and the physical realities of the site, rather than solely improving software. To transition from "hierarchical guessing" to "collaborative knowing," the following actions are recommended.

8.1 Initial 20% Phase

8.1.1 The Critical First 20% Phase

Research indicates that the initial 20% of a project represents the most critical period for identifying risks before they become physically irreversible. To address this, organizations should establish a mandatory "20% Reality Check" milestone. Instead of a conventional progress meeting, this milestone should take the form of a "Calibration Workshop" in which the project is audited from two distinct perspectives simultaneously.

During this workshop, junior team members are responsible for validating the technical "ground truth" by confirming the accuracy of materials, soil conditions, and site calculations. Senior professionals concurrently focus on the strategic "shield," ensuring that legal permits, budgets, and political compliance are adequately addressed. Advancement to the subsequent project stage should occur only when both groups reach consensus. This approach mitigates the risk of the "Illusion of Progress" and helps prevent the project from surpassing the point of no return with unresolved foundational issues.

8.2 Overall

8.2.1 The Two-Way Team Model

To address the divide between experienced professionals and newcomers, organizations should move from traditional top-down hierarchies to a Two-Way Team Model. This approach acknowledges that senior staff possess a comprehensive understanding of organizational objectives but may overlook minor technical errors due to familiarity. In contrast, junior staff are attentive to technical details but often lack the experience required to navigate complex regulatory environments.

Within this framework, junior project managers are authorized to perform detailed technical audits, referred to as 'Micro-Checks,' on the work of senior colleagues. Granting juniors the authority to identify even minor discrepancies, such as a 1mm measurement error, enables the team to address small issues before they develop into significant problems. In exchange, senior staff serve as mentors and provide legal and regulatory guidance. Their responsibilities expand beyond task delegation to include instructing juniors on interactions with government officials, interpreting contracts, and managing political risks not typically covered in formal education.

8.2.2 Reverse Training and Geographic Intelligence

Research indicates that a project manager’s environment functions as a "silent teacher," fostering distinct strengths across regions. To leverage these differences, a "Reverse Training" philosophy is proposed. This approach challenges the traditional assumption of Western management superiority and instead posits that diverse cultures offer valuable lessons to one another.

In the German context, emphasis should be placed on "Chaos Adaptability." German project managers often rely extensively on digital tools, which can lead to decision paralysis when plans fail. Implementing "Crisis Simulations" inspired by the Indian concept of Jugaad, where digital tools are removed, would encourage managers to adapt to the environment rather than depend solely on spreadsheets. Conversely, in the Indian context, the focus should be on "Digital Standardization." While Indian project managers excel in adaptability and human-based trust, they may encounter challenges with scaling and auditing. Adopting the German "Process Shield" would facilitate a transition from "relationship-based trust" to "data-based trust," thereby supporting alignment with international professional standards.

8.2.3 Transitioning to Truth-Link Technology

The "Trust Deficit" remains a significant challenge, as 93.3% of managers bypass digital dashboards due to concerns that "paper lies." To address this issue, organizations should transition toward "Truth-Link" Socio-Technical Systems. Rather than relying on manual reports, which many project managers regard as "polite fictions," embedding live sensors directly into the physical project is recommended.

Utilizing technologies such as acoustic sensors to monitor machine health or foundation sensors to assess concrete curing enables the creation of a "Digital Twin" that delivers high-fidelity, real-time data. This approach addresses project managers’ psychological need for "Somatic Truth," defined as the ability to directly observe, hear, or sense project progress. Access to live data via mobile devices reduces the necessity for frequent site visits solely to verify report accuracy. Consequently, the industry shifts from "Manual Checking" to a more efficient practice of "Data-Driven Feeling."

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10 Annex

The following interview questionnaire were asked as a structured tool to gather insights from participants related to the research topic. The information collected contributes to understanding key variables and strengthening the study outcomes.

1. Short Background

- Which industry are you mainly working in right now? (e.g., construction, IT, manufacturing, other)
- How many years have you worked as a project manager?
- Roughly how many projects have you managed from start to finish?

2. Early-Stage Risk Focus

- When you start a new project, what are the first three risks that come to your mind? Please just list them for a typical project in your current job.
- Regarding these three risks: Are they mostly inside the project (directly controllable) or outside the project (uncontrollable factors such as politics, regulations, market, client decisions)?

3. Risk Identification in the First 20% of the Project

- In the first 20% of a typical project, what do you actually do to identify risks? Could you walk me through the concrete steps you take?
- Do you usually use any checklists or templates? If yes, which ones?
- Do you talk to specific people to find risks? Who exactly?
- Do you look at documents from previous projects (lessons learned, old risk lists)?
- Honestly reflecting on your own practice, what are the two or three methods you use most often to find risks at the beginning of a project?

4. Intuition vs Tools

- At the beginning of a project, when deciding which risks to take seriously, what matters more for you personally:
- Your own experience and gut feeling
- Or formal tools and methods (e.g., risk matrices, registers, templates)?
- If you had to choose, would you say you mainly rely on:
    - Mostly experience and intuition
    - A balance of intuition and tools
    - Mostly formal tools and methods

5. Decision-Making with Incomplete Information

- Can you describe one concrete example where, at the start of a project, you had to make an important decision but did not have all the information you wanted?
    - What information was missing?

    - What did you decide to do?
    - Why did you decide that way?
- If you faced the same situation today, would you decide the same or differently? Why?

6. Time and Management Pressure

- When there is strong time pressure at the start of a project, what changes in the way you handle risks, compared to when you have comfortable time?
- When top management pushes to ‘start quickly’ or ‘show progress fast’, how does that influence which risks you look at and which you ignore at the beginning?

7. Cross-Industry Habits and Blind Spots

- In your industry, what are two or three typical risks that people from other industries often underestimate or do not see at all?
- If a project manager from a different industry joined your project tomorrow, what would you warn them about in the early planning phase?
- Have you ever worked in another industry? If yes, what habits about risk from your old industry did not work well in the new one?

8. Practical Outcomes: Supporting Project Managers

- From your point of view, what two or three things would help project managers in your industry identify risks earlier and make better decisions at the start of a project?
- If you could design a short training for project managers coming from another industry into your field, what are the top two or three risk topics they must learn to avoid typical mistakes?